Asymptotic flatness of subset-sum landscapes of primes: the sub-peak spectrum and the constant $\sqrt{3}/2$

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Abstract

Let B be a finite set of odd primes and $r_B(m)$ the number of subsets of B summing to m. The companion paper reduces the asymptotic behaviour of the subset-sum landscape of B to a single quantity, the *flatness* ε_d, which measures the pointwise oscillation of r_B on a bounded window. This paper develops the Fourier-side input needed to bound ε_d: we determine the complete spectrum of sub-peaks of the characteristic product F_B, we prove that modulus 6 is the unique maximiser of the per-element sub-peak height, and we show that for a truncated set of primes an entire family of sub-peaks vanishes *identically*. Concretely, the peak at modulus $q = 2m$ with m odd is exactly 0 whenever $m \in B$; for B a set of primes this kills every modulus $2p$ with $p \in B$, so that the only competitors of the modulus-6 peak are the modulus-4 floor $(1/\sqrt{2})^{|B|}$, which never vanishes, and a finite list depending on the truncation level. We also derive and verify numerically a second-order formula for the modulus-6 ripple, giving amplitude and phase to within 0.1% over $k = 18, \ldots, 76$. Combining these with the deep-minor bound of Section 9, which is proved here, we obtain $\varepsilon_d(k) \to 0$ at a geometric rate of at most $\sqrt{3}/2$ per element, and hence $\mathrm{lm}_P(n)/\deg_P(n) \to \Gamma(P) = 5.34920\ldots$ for the odd primes; this answers Problem 10.1 of the companion paper.

Two things about the status of that theorem, stated here rather than left to be discovered. *It carries no hypothesis*: earlier versions assumed the deep-minor bound, and it is now Proposition 9.3. But no hypothesis is not the same as written out in full, and two steps of that proposition are not yet written to referee standard — the substitution of the quoted exponential-sum bound in step 2, and the excision in step 4 of Section 9. Both are flagged where they occur and tracked in Section 11; neither is believed to need a new idea, and until they are written the theorem should be read as a complete argument with two steps still owed to the reader. *It is also ineffective*, at exactly one point: the Siegel–Walfisz constant, which Appendix 12 isolates as the only ineffective constant in the paper. Key steps are formally verified in Lean 4.

1 Introduction

1.1 The quantity to be bounded

For a finite set B of odd positive integers write $r_B(m) = \#\{S \subseteq B : \sum S = m\}$, $T = \sum B$, and

$$F_B(\theta) = \prod_{a \in B} (1 + e^{ia\theta}), \qquad r_B(m) = \frac{1}{2\pi} \int_0^{2\pi} F_B(\theta) e^{-im\theta} \, d\theta.$$

The companion paper shows that the ratio of the number of strict local minima to the number of ground states of the subset-sum landscape converges to an explicit arithmetic invariant

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provided the *flatness*

$$\varepsilon_d = \max_{m,m' \in \mathrm{Win}_d} \left| \frac{r_{B_d}(m)}{r_{B_d}(m')} - 1 \right|$$

tends to 0, where B_d is the subsequence of elements exceeding $2d$ and Win_d is a bounded window about $T/2$. Bounding ε_d is a circle-method problem: the main arc at $\theta \approx 0$ gives a smooth Gaussian term, and every deviation from smoothness is carried by the sub-peaks of $|F_B|$ at the rationals $\theta = 2\pi j/q$.

1.2 Results

Section 2 determines the sub-peak spectrum. Writing $G_B(\theta) = \prod_{a \in B} \cos(a\theta/2)$, so that $|F_B| = 2^b |G_B|$, where $b = |B|$, the natural per-element height at modulus q is

$$M(q) = \Big(\prod_{\substack{1 \le r \le q \\ \gcd(r,q)=1}} |\cos(\pi r/q)| \Big)^{1/\varphi(q)}.$$

Proposition 2.1 evaluates $M(q)$ in closed form, and Theorem 2.3 shows that $\sqrt{3}/2$ is its supremum, attained only at $q = 6$.

The novelty of Section 3 is that $M(q)$, being an upper bound valid for equidistributed sequences, is far from what a set of primes actually realises. Lemma 3.1 shows the peak at $q = 2m$ (m odd) is *exactly zero* as soon as $m \in B$; Corollary 3.2 converts this into the surviving spectrum of a truncated prime set. The consequence for the circle method is that the minor arcs carry, up to a fixed finite list, only two live peaks.

Section 4 derives a second-order formula for the modulus-6 ripple, with amplitude correction K_2 and phase drift ψ; Section 5 treats the main arc and identifies the first correction to the central density, resolving an unexplained constant of the companion paper. Section 6 reports the numerical verification, Section 8 makes the finite list of remaining moduli effective, Section 9 treats the deep minor arcs, and Section 10 assembles the main theorem.

1.3 Relation to the literature

Three bodies of work are close enough to require comment.

Oscillations in partition counts. Murthy and Brack [MB26] give a semiclassical, periodic-orbit account of oscillations in partition counting functions; conceptually this is the same mechanism as ours, a few isolated saddle contributions producing a visible ripple. Their detailed oscillation analysis is for partitions into distinct *squares*, where the relevant orbits are indexed by Pythagorean triples, and their treatment of prime partitions (their §7) is asymptotic rather than oscillatory. The limit also differs: they let the integer being partitioned tend to infinity with the part set fixed, whereas we let the *number of parts k* tend to infinity. The dictionary is one line: their periodic orbits are our sub-peaks at rational $\theta = 2\pi j/q$. We would add that the question they leave open — why regular oscillations survive only for certain special sequences — has, in the subset-sum setting, the exact answer given by Lemma 3.1 and Corollary 3.2: an oscillation at modulus q survives precisely when q avoids the cyclotomic vanishing $q = 2m$ with $m \in B$, and the surviving heights are the $M(q)$ of Proposition 2.1.

Smoothness of partition functions. Bateman and Erdős [BE56] prove $p_A(n-1)/p_A(n) \to 1$ for a *fixed infinite* set A of parts with *repetitions allowed.* Our situation differs in both respects: our part set is a finite truncation that grows with k, and parts are used at most once, so the generating function is $\prod_{a \in B}(1 + x^a)$ rather than $\prod_{a \in A}(1 - x^a)^{-1}$. We do not see how to deduce the finite-truncation statement from theirs. The same remark applies to Roth and Szekeres [RS54] and to Vaughan [V08], both of which treat the fixed infinite set of all primes.

Landscapes of the number partitioning problem. Ferreira and Fontanari [FF98] compute the fraction of metastable states — that is, $\mathrm{lm}/2^k$ — analytically for uniform random weights; Alyahya and Rowe [AR14] give a formula in terms of the coefficient of variation; Stadler, Hordijk and Fontanari [SHF03] analyse barrier trees. All of these average over *random* weights, whereas the companion paper and this one treat a *specific* sequence and an order-sensitive invariant.

Finally, the anti-concentration literature (Littlewood–Offord and its inverse forms) bounds the *maximum* of r_B; what we need is a two-sided bound on the *ratio* at nearby points, which does not follow from a one-sided bound.

1.4 Formal verification

Statements marked `name` are machine-checked in Lean 4 against Mathlib; the repository builds with no `sorry` and no additional axioms. The canon currently consists of 97 theorems.

Remark 1.1 (A convention about the remarks). Several approaches to the results below were tried and abandoned. Where an abandoned approach is instructive — because the obstruction it meets is real, or because the natural first guess is false — we record it as a remark with the reason, rather than silently omitting it. Readers who want only the proofs may skip those; readers who want to know where the ground is soft should read them first.

2 The sub-peak spectrum

The two statements of this section are classical: Proposition 2.1 is the resultant $\mathrm{Res}(\Phi_q, x+1)$ written out, and Lemma 2.2 is a standard exercise. We record both with proofs because we use the precise constants, and because the interpretation of $M(q)$ as the per-element height of a sub-peak of a subset-sum generating function seems not to have been made before.

Proposition 2.1 (Cyclotomic evaluation; classical)**.** *For every integer $q \geq 3$,*

$$M(q) = \tfrac{1}{2}\,|\Phi_q(-1)|^{1/\varphi(q)},$$

where Φ_q is the q-th cyclotomic polynomial.

Proof. Put $\zeta = e^{2\pi i/q}$. For $1 \leq r \leq q-1$ we have $|1+\zeta^r| = 2|\cos(\pi r/q)|$. Since $\Phi_q(x) = \prod_{\gcd(r,q)=1}(x-\zeta^r)$, evaluating at $x=-1$ gives $\prod_{\gcd(r,q)=1}|1+\zeta^r| = |\Phi_q(-1)|$, hence $\prod_{\gcd(r,q)=1}|\cos(\pi r/q)| = 2^{-\varphi(q)}|\Phi_q(-1)|$. Take $\varphi(q)$-th roots. □

Lemma 2.2 (Classical values)**.** *For $q \geq 3$ one has $\Phi_q(-1) = p$ if $q = 2p^j$ with p an odd prime and $j \geq 1$; $\Phi_q(-1) = 2$ if $q = 2^j$ with $j \geq 2$; and $\Phi_q(-1) = 1$ otherwise.*

Proof. We use three standard identities: $\Phi_n(1) = p$ if $n = p^j$ is a prime power and $\Phi_n(1) = 1$ if n has at least two distinct prime factors; for odd $n \geq 3$, $\Phi_{2n}(x) = \Phi_n(-x)$; and for even m, $\Phi_{2m}(x) = \Phi_m(x^2)$. If q is odd then $\Phi_q(-1) = \Phi_{2q}(1) = 1$ because $2q$ has two distinct prime factors. If $q = 2m$ with m odd then $\Phi_q(-1) = \Phi_m(1)$, which is p when $m = p^j$ and 1 otherwise. If $4 \mid q$, write $q = 2m$ with m even; then $\Phi_q(-1) = \Phi_m(1)$, which equals 2 exactly when m is a power of 2, i.e. when $q = 2^j$, and equals 1 otherwise. □

Theorem 2.3 (Extremality of modulus six)**.** $\sup_{q\geq 3} M(q) = \sqrt{3}/2$*, attained only at $q = 6$. The second largest value is $M(10) = 5^{1/4}/2 \approx 0.7477$, and $M(q) \leq 1/\sqrt{2}$ for every $q \notin \{6, 10\}$.*

Proof. By Proposition 2.1 and Lemma 2.2, $M(q) = 1/2$ unless $q = 2p^j$ or $q = 2^j$. For $q = 2^j$ ($j \geq 2$) we get $M = \frac{1}{2} \cdot 2^{1/2^{j-1}}$, which is decreasing in j with maximum $1/\sqrt{2}$ at $q = 4$. For $q = 2p^j$ we get $M = \frac{1}{2}\, p^{1/(p^{j-1}(p-1))}$; the exponent decreases in j, so the maximum over the family occurs at $j = 1$, where $M = \frac{1}{2}p^{1/(p-1)}$. The map $p \mapsto (\log p)/(p-1)$ is strictly decreasing for $p \geq 3$, since its derivative has the sign of $(p-1)/p - \log p < 0$ (as $\log p \geq \log 3 > 1 > (p-1)/p$). Hence the values in this family are $\sqrt{3}/2 > 5^{1/4}/2 > 7^{1/6}/2 \approx 0.6916 > \cdots$, and for $j \geq 2$ the largest is $q = 18$ with $\frac{1}{2}3^{1/6} \approx 0.6005$. Combining the three families gives the statement. $\square$

(Lean: `M12_six_maximal` and `M12_gap` verify the finite comparison at the twelfth power, where all the relevant values are rational.)

3 The vanishing family and the surviving spectrum

Lemma 3.1 (Exact vanishing). *Let $q = 2m$ with m odd. If $m \in B$ then $F_B(e^{2\pi i/q}) = 0$; equivalently $G_B(2\pi/q) = 0$.*

Proof. The factor of F_B indexed by m is $1 + e^{i\pi} = 0$. $\square$

(Lean: `F_eq_zero_of_mem` for the product form, `F_eq_zero_at_two_pi_div` for the statement at $\theta = 2\pi/q$, `F_nat_eq_zero_of_mem` for integer sequences, and `not_mem_of_F_ne_zero` for the contrapositive. The special case $q = 6$, $m = 3$ is `normSq_at_pi` of the companion paper.)

The point of Lemma 3.1 is its converse for prime sequences: the residue class $a \equiv m \pmod{2m}$ consists of odd multiples of m, so if B is a set of primes and $m > 1$ then B meets that class if and only if m itself lies in B. This gives a complete description.

Corollary 3.2 (Surviving spectrum). *Let B_d be the set of primes in $(2d, x]$ and $b = |B_d|$. Then:*

1. $|G_{B_d}(2\pi/6)| = (\sqrt{3}/2)^b$ exactly, *provided $d \geq 2$ so that $3 \notin B_d$; no equidistribution is needed, because both classes $\pm 1 \pmod 6$ give $|\cos(\pi a/6)| = \sqrt{3}/2$.*
2. $|G_{B_d}(2\pi/4)| = (1/\sqrt{2})^b$ exactly, *for every odd sequence whatsoever, since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for all odd a. This is the Fourier form of the modulus-4 obstruction of the companion paper.*
3. *$|G_{B_d}(2\pi/(2p))| = 0$ exactly, for every prime p with $2d < p \leq x$. The vanishing is* pointwise *and must not be read as the absence of a peak: one factor is removed at one point, and the surrounding mass is still $(M(2p) + o(1))^b$. The two maxima flanking the resulting notch sit at height $(M(2p) + o(1))^b b^{-3/2}$, the notch having width $\asymp 1/p$ — see Remark 11.20, where this is measured and matched against Lemma 11.15.*
4. *For the finitely many primes $p \leq 2d$ the peak at $q = 2p$ survives, at height $(M(2p) + o(1))^b \leq (5^{1/4}/2 + o(1))^b$ by the prime number theorem in arithmetic progressions.*
5. *For $q = 2m$ with m odd composite, and for $q = 2^j$ with $j \geq 3$, the height is $\leq (M(q) + o(1))^b \leq (\frac{1}{2}3^{1/6} + o(1))^b$.*
6. *For q odd the height is $\leq (\frac{1}{2} + o(1))^b$, and at $\theta = \pi$ it is 0 by parity.*

[mixed by item, and the mixture is the point. Items (i)–(iii) and (vi) are *exact identities*, proved in place, using no equidistribution whatsoever. Items (iv)–(v) do use equidistribution, and only for a finite list of moduli, for which effective error terms are available — these are the two items made effective in Proposition 8.1. A reader who takes the whole corollary as equidistribution-dependent will overstate what the paper assumes; one who takes it all as exact will understate it.]

Remark 3.3. Items (i)–(iii) and (vi) are exact identities; only (iv)–(v) use equidistribution, and only for a finite list of moduli, for which effective error terms in the prime number theorem in arithmetic progressions are available. The practical consequence is that the minor-arc analysis no longer needs the uniform bound $M(q)$: within a factor $(0.75/0.866)^b$ of the modulus-6 peak there are, for fixed d, only $q = 4$ and the finite list $\{2p : p \leq 2d\}$.

Figure 2 shows the measured spectrum for the first 24 primes ≥ 5; the vanishing predicted by Lemma 3.1 is visible at $q = 10, 14, 22$, where the computed values are $6.0 \cdot 10^{-20}$, $9.0 \cdot 10^{-21}$ and $2.5 \cdot 10^{-22}$, i.e. numerically zero, against predicted heights $9.3 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$ and $6.0 \cdot 10^{-8}$ under equidistribution.

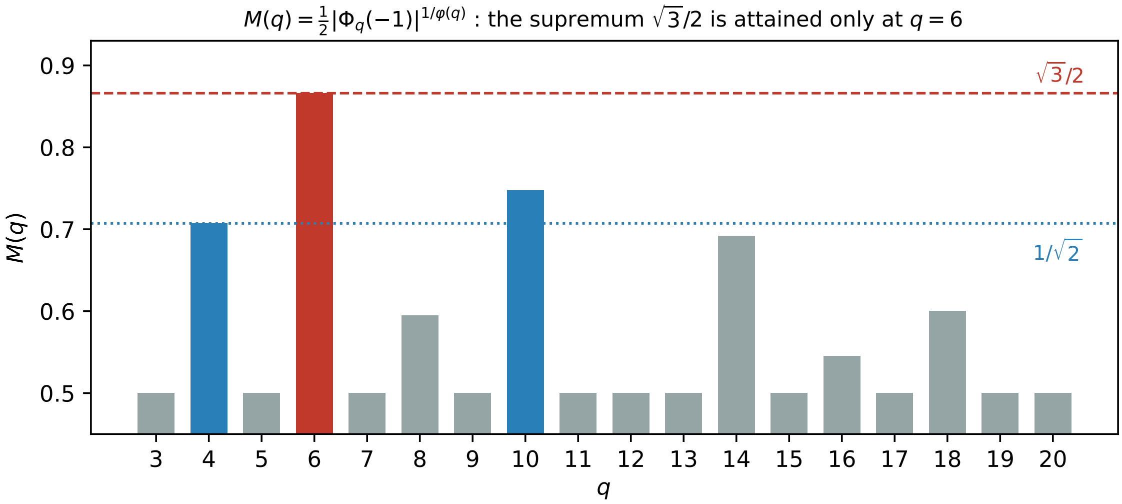

Figure 1: The per-element sub-peak height $M(q)$ (exact values, by Proposition 2.1). The supremum $\sqrt{3}/2$ is attained only at $q = 6$; $q = 10$ and $q = 4$ are the runners-up.

4 The modulus-six ripple to second order

Throughout this section B is a set of b odd primes with $3 \notin B$. Put

$$S_2 = \sum_{a \in B} a^2, \quad S_4 = \sum_{a \in B} a^4, \quad \Delta = \sum_{a \in B} \tau_a a, \quad \Delta_3 = \sum_{a \in B} \tau_a a^3,$$

where $\tau_a = +1$ if $a \equiv 1 \pmod 6$ and $\tau_a = -1$ if $a \equiv 5 \pmod 6$, and let c_1, c_5 be the corresponding class counts. Write $\mathrm{Main}(m)$ for the Gaussian main-arc term and $\nu = T/2 - m$.

Lemma 4.1 (Local data at $\theta = \pi/3$)**.** *Every $a \in B$ satisfies $\cos^2(a\pi/6) = 3/4$, $\tan(a\pi/6) = \tau_a/\sqrt{3}$ and $\sec^2(a\pi/6) = 4/3$. Consequently, with $h(t) = \log|G_B(\pi/3 + t)|$,*

$$h'(0) = -\frac{\Delta}{2\sqrt{3}}, \quad h''(0) = -\frac{S_2}{3} =: -V_6, \quad h'''(0) = -\frac{\Delta_3}{3\sqrt{3}}, \quad h''''(0) = -\frac{S_4}{3}.$$

Moreover $\frac{d}{d\theta} \arg(1 + e^{ia\theta}) = a/2$ exactly, so the phase of F_B is linear along the arc and all curvature lives in $|G_B|$.

(Lean: `cos_sq_pi_mul_div_six_of_one`, `cos_sq_pi_mul_div_six_of_five`, `sec_sq_pi_mul_div_six`.)

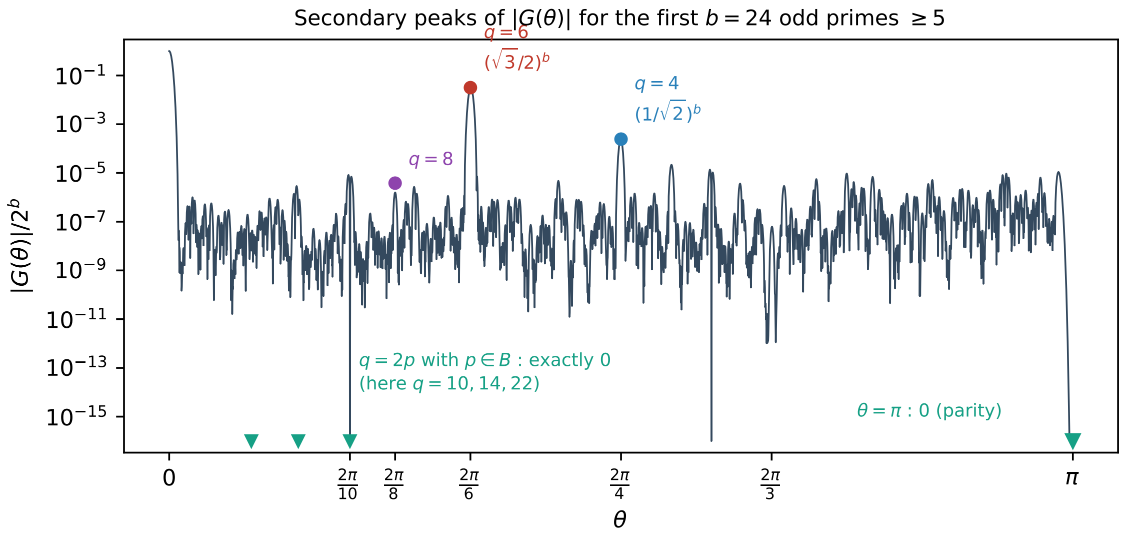

Figure 2: Measured $|G_B(\theta)|/2^b$ for the first $b = 24$ primes ≥ 5. The peaks at $q = 2p$ with $p \in B$ are identically zero (Lemma 3.1, proved); $\theta = \pi$ vanishes by parity. Computed exactly; no asymptotics enter this figure.

Proposition 4.2 (Second-order ripple formula)**.** *For m in a bounded window about $T/2$,*

$$\frac{r_B(m)}{\mathrm{Main}(m)} - 1 = 2\Big(\frac{\sqrt{3}}{2}\Big)^{b+1} K_2 \cos\Big(\frac{\pi}{6}(2m - (c_1 - c_5)) + \psi(\nu)\Big) + E,$$

where

$$K_2 = \exp\Big[\frac{\Delta^2}{8S_2} - \frac{3}{8}\frac{S_4}{S_2^2} + \frac{\Delta\Delta_3}{4S_2^2} + \frac{5\Delta_3^2}{24S_2^3} - \frac{3S_4\Delta^2}{16S_2^3}\Big], \qquad \psi(\nu) = -\frac{\sqrt{3}}{2}\,\nu\,\frac{\Delta + \Delta_3/S_2}{S_2}. \quad (1)$$

[derived, with the remainder *measured* rather than bounded — said here at the display, because the two are easy to confuse. The equality is obtained by Laplace's method at the saddle and the derivation is given below; the omitted cumulant contributions are named in Remark 4.4; but E is controlled numerically on the tested range (Table 1, Remark 6.1) and *not* by a proved asymptotic bound. This is therefore an exact derivation with a numerically controlled remainder, not a theorem with a proved error term. Theorem 10.1 inherits the distinction: its conclusion $\varepsilon_d \to 0$ does not use K_2 beyond the unconditional bound of Remark 10.3, while its sharp constant $\sqrt{3}/2$ does.]

Derivation. Laplace's method at the saddle $t^* = (L + i\nu)/V_6$ with $L = h'(0)$: the exponent contributes $(L + i\nu)^2/(2V_6) + C_3(L + i\nu)^3/(6V_6^3) + \cdots$, the Gaussian-fluctuation factor $-\frac{1}{2}\log(-h''(t^*))$ contributes $C_3 t^*/(2V_6) + C_4(t^*)^2/(4V_6) + \cdots$, and the standard constant is $5C_3^2/(24V_6^3) + C_4/(8V_6^2)$. Substituting $C_3 = -\Delta_3/(3\sqrt{3})$, $C_4 = -S_4/3$ and $V_6 = S_2/3$ from Lemma 4.1, and separating real from imaginary parts, gives K_2 and ψ. The width factor $\sqrt{V_0/V_6} = \sqrt{3}/2$ accounts for the exponent $b + 1$. $\square$

Remark 4.3 (The two branches)*.* The term $\Delta^2/(8S_2)$ is positive and driven by the prime race between the classes $\pm 1 \bmod 6$; the term $-\frac{3}{8}S_4/S_2^2 \asymp -9/(8b)$ is negative and class-independent. Their competition produces the two branches seen in the data. In the measured range $c_1 - c_5 \in \{-1, -3\}$, and the branch is decided by that value: $c_1 - c_5 = -3$ forces $|\Delta| \geq 135$ and $K_2 > 1$, while $c_1 - c_5 = -1$ keeps $|\Delta| \leq 47$ and $K_2 < 1$.

Remark 4.4 (Error policy). The omitted contributions are fifth- and sixth-cumulant terms and quartic crosses in Δ, of order $O(|\Delta_5|/S_2^{5/2} + S_6/S_2^3 + S_4\Delta^4/S_2^4 + |\Delta_3|^3/S_2^{9/2})$ per unit of $K_2 - 1$. We state the remainder as an explicit numerical bound on the verified range together with $o(1)$ asymptotics, rather than adding further terms, which carry no structural information.

5 The main arc and the central density

The main arc contributes the smooth part of r_B. The leading term is the local central limit theorem; what matters for the companion paper's absolute formula is the *first correction* to it.

Proposition 5.1 (Central density with first correction). *Let B be a set of b odd integers, $V_0 = S_2/4$, and let m be at the centre $T/2$. Then*

$$r_B(m) = \frac{2^b}{\sqrt{2\pi V_0}}\Big(1 - \frac{S_4}{4S_2^2} + O\big((S_4/S_2^2)^2\big)\Big).$$

[proved. The derivation below is labelled *Derivation* because it is a Taylor and Laplace computation rather than a formal argument, but every term is written out and the O-constant is explicit; the only step taken as standard is extending the main-arc integral to $\mathbb{R}$. Said here because Proposition 4.2 carries the same label and a *different* status — there the remainder is measured, here it is bounded.]

Derivation. Write $\log G_B(\theta) = \sum_{a\in B}\log\cos(a\theta/2) = -\frac{V_0}{2}\theta^2 - \frac{S_4}{192}\theta^4 + O(S_6\theta^6)$, using $\log\cos u = -u^2/2 - u^4/12 + O(u^6)$ and $u = a\theta/2$. Since G_B is even (Lean: `G_even`), all odd-order terms vanish identically, so the first correction is the quartic one. Integrating $e^{-V_0\theta^2/2}(1 - \frac{S_4}{192}\theta^4)$ over the main arc and using $\int \theta^4 e^{-V_0\theta^2/2} = 3\sqrt{2\pi}V_0^{-5/2}$ gives the stated factor $1 - \frac{S_4}{192}\cdot 3V_0^{-2} = 1 - \frac{S_4}{4S_2^2}$. □

Remark 5.2 (The absolute formula of the companion paper). The companion paper predicts the number of strict local minima by $\mathrm{lm} \approx \Gamma\cdot 2^b/\sqrt{2\pi V_0}$, i.e. by replacing the ground-state count by its Gaussian estimate. The measured ratio of the two sides sits systematically below 1 — about 0.97 — with no explanation offered there. Proposition 5.1 identifies the deficit as $S_4/(4S_2^2)$, which at $k = 20$ equals 0.0246. Section 6.4 verifies this. We warn against one confusion: the value 0.972 appearing here (the ratio lm/pred at $k = 20$) is *not* the 0.972 in the companion paper's list of $Q(k)$, which is $Q(18)$. Those are different quantities and the numerical coincidence is accidental: Q is formed with the *measured* ground-state count, so it carries no Edgeworth correction, and it oscillates about 1 rather than sitting below it.

6 Numerical verification

All computations below are exact integer dynamic programming for r_B, followed by a discrete Fourier extraction; logs are included in the repository.

6.1 The amplitude correction

Table 1 compares the measured amplitude ratio against Proposition 4.2 with and without the two second-order terms.

Δ_3^2 coefficient	$S_4\Delta^2$ term	branch A mean	branch B mean
45/24	no	0.00061	0.00736
45/24	yes	0.00060	0.00400
5/24	no	0.00073	0.00411
5/24	yes	0.00072	**0.00073**

Table 1: Mean of $|\text{measured}/K_2 - 1|$ over $k = 18, \ldots, 40$. Both corrections are needed: either one alone leaves a residual of about 0.4% on branch B.

Remark 6.1 (How far the check reaches, and what stops it). The comparison above was originally made over $k = 18, \ldots, 40$. We have extended it, computing r_B exactly in integer arithmetic, to $k \le 100$:

range	mean $\lvert\text{meas}/K_2 - 1\rvert$	max	n
$k = 18, \ldots, 40$	0.00073	0.00339	12
$k = 44, \ldots, 76$	0.00077	0.00205	9
$k = 80, \ldots, 100$	0.0102	0.0211	6

So the agreement is unchanged out to $k = 76$ — the range of the claim roughly doubles at no cost — and then degrades. The degradation is a limit of the measurement, not of the formula, and one can say so quantitatively. The ripple being extracted has relative size $4(\sqrt{3}/2)^b$, which is $1.5 \cdot 10^{-2}$ at $k = 40$ but $2.6 \cdot 10^{-6}$ at $k = 100$, whereas the smooth background it sits on — the Edgeworth correction, of size S_4/S_2^2 — only decays like $1/b$, from 0.051 to 0.020. The ratio of background to signal therefore climbs from 3.5 at $k = 40$ to $7.6 \cdot 10^3$ at $k = 100$. Multiplying the observed discrepancy by the reciprocal of that ratio gives $8.9, 6.5, 4.1, 2.8$ (all $\times 10^{-6}$) at $k = 64, 76, 88, 100$: *constant to within a factor of three while the ratio itself moves by two orders of magnitude.* In other words the leakage of the smooth background into the modulus-6 Fourier bin is a fixed $\approx 5 \cdot 10^{-6}$ of that background, and it becomes visible in the ratio only because the signal is shrinking underneath it. Below $k \approx 52$ the residual is instead dominated by the accuracy of K_2 itself, at 3–$6 \cdot 10^{-4}$. Reaching beyond $k \approx 80$ would require subtracting the background to better than one part in 10^5, which is a question about the reference expansion and not about Proposition 4.2.

On branch B the individual values of $\text{measured}/K_2$ are 0.9991 ($k = 32$), 0.9990 ($k = 34$) and 0.9997 ($k = 40$), against 0.9904, 0.9923 and 0.9952 without the corrections. The largest residual over the whole range is 0.0034, attained at $k = 18$. The omitted contributions are dominated by the class-independent tail $O((S_4/S_2^2)^2)$, and S_4/S_2^2 decreases monotonically over the range, from 0.1111 at $k = 18$ to 0.0507 at $k = 40$; hence the residual is largest at the smallest k. That the Δ-dependent quantity Δ_3^2/S_2^3 is in fact largest at $k = 32$ ($2.70 \cdot 10^{-3}$, against $1.04 \cdot 10^{-3}$ at $k = 18$) without producing an excess residual there is corroborating evidence that the Δ-dependence is already absorbed into K_2.

6.2 The phase drift

Fitting the measured phase slope to $\alpha\Delta/S_2 + \beta(c_1 - c_5)/S_2 + \gamma/S_2$ over $k = 18, \ldots, 40$, only α is significant, at every window width tested: $t(\alpha)$ ranges from $+4.26$ to $+5.16$, while $|t(\beta)|$ and $|t(\gamma)|$ never exceed 1.6. The one-parameter model measured $= c \cdot \psi'$ gives $c = 0.884, 0.888, 0.904$ and 0.970 at half-widths $24, 48, 96$ and 192, with R^2 from 0.978 to 0.986; restricting to $|\Delta| \ge 17$ raises R^2 to 0.993. We conclude that ψ as stated is correct, with the deficit in c a finite-window effect.

Remark 6.2 (Extraction). The amplitude is extracted from a narrow window of fixed half-width 24, i.e. eight full periods of the modulus-6 ripple, which over $k = 18, \dots, 40$ is between 0.08 and 0.30 standard deviations; both requirements — enough periods, and narrower than σ — hold throughout. Wide windows corrupt the Gaussian reference. As a check, at $k = 18$ the ratio measured/K_2 is 0.9988, 0.9974, 1.0034 and 1.0065 at half-widths $12, 16, 24$ and 32, so the window choice moves the residual by less than 1%. The phase slope requires *wide* windows, half-width 96 to 192; at half-width 384 the normalisation breaks down in the Gaussian tails. For $|\Delta| \leq 3$ the true slope is at the level of extraction noise in narrow windows.

6.3 The surviving spectrum

Table 2 verifies Corollary 3.2 by varying the truncation level d: a peak at $q = 2p$ is dark exactly when p is still present in B_d, and lights up as soon as the truncation removes p. All 50 predictions tested were correct, and $q = 6$ was the tallest surviving peak in every case.

d	removed primes	$q = 6$	$q = 10$	$q = 14$	tallest
1	—	0	0	0	$q = 4$
2	3	$3.66 \cdot 10^{-2}$	0	0	$q = 6$
3	$3, 5$	$4.22 \cdot 10^{-2}$	$1.03 \cdot 10^{-3}$	0	$q = 6$
4	$3, 5, 7$	$4.88 \cdot 10^{-2}$	$1.75 \cdot 10^{-3}$	$4.33 \cdot 10^{-4}$	$q = 6$

Table 2: $|G_{B_d}(2\pi/q)|/2^b$ for A the first 24 odd primes. Entries equal to 0 are exact zeros.

Finally, at $q = 4$ and $q = 6$ the measured heights agree with $(1/\sqrt{2})^b$ and $(\sqrt{3}/2)^b$ to six decimal places for every d, confirming that no equidistribution hypothesis enters items (i) and (ii) of Corollary 3.2. For the remaining moduli the ratio to $M(q)^b$ lies between 0.14 and 2.3: the equidistribution approximation is *not* accurate at these sizes, which is precisely why the exact identities matter. This is to be expected, since the error in the prime number theorem in arithmetic progressions decreases only logarithmically, so at $b = 21, \dots, 31$ the $o(1)$ in items (iv)–(v) of Corollary 3.2 has not yet become small.

6.4 The central density

We test Proposition 5.1 using only ground-state counts. Write $\lambda(B) = \deg(B)\sqrt{2\pi V_0}/2^b - 1$ for the measured relative deficit, where $\deg(B)$ is the number of subsets attaining the minimum of $|\sum S - \lfloor T/2 \rfloor|$. Proposition 5.1 predicts $\lambda \approx -S_4/(4S_2^2)$.

k	deg	measured λ	predicted	difference
16	394	-0.024764	-0.031338	$+0.006573$
18	1290	-0.025573	-0.027754	$+0.002181$
20	4344	-0.024036	-0.024629	$+0.000594$
24	51068	-0.021467	-0.020830	-0.000637
28	634568	-0.017663	-0.017574	-0.000090

Table 3: Central-density deficit for the first k odd primes. The agreement is within 0.5% for $k \geq 18$ and improves to 0.01% at $k = 28$.

Over an ensemble of 100 random odd sequences per size (drawn from the same range as the primes), the mean absolute difference is 0.0171 at $k = 16$, 0.0079 at $k = 18$ and 0.0038 at $k = 20$; the ensemble is noisier than the primes because deg is a small integer subject to arithmetic fluctuation.

The consequence for Remark 5.2 is direct. Writing $\rho = \mathrm{lm}/(\Gamma \cdot 2^b/\sqrt{2\pi V_0})$ for the ratio in the absolute formula, the ensemble mean of ρ is 0.9462, 0.9621, 0.9672 and 0.9716 at $k = 12, 16, 18, 20$, while the mean of $\rho/(1 - S_4/(4S_2^2))$ is 0.9812, 0.9888, 0.9912 and 0.9933. The correction removes about three quarters of the deficit, and the share it removes grows with k. We therefore record the systematic part of the 0.97 as explained; the residual 0.7% at $k = 20$ is a separate and smaller effect, still decreasing.

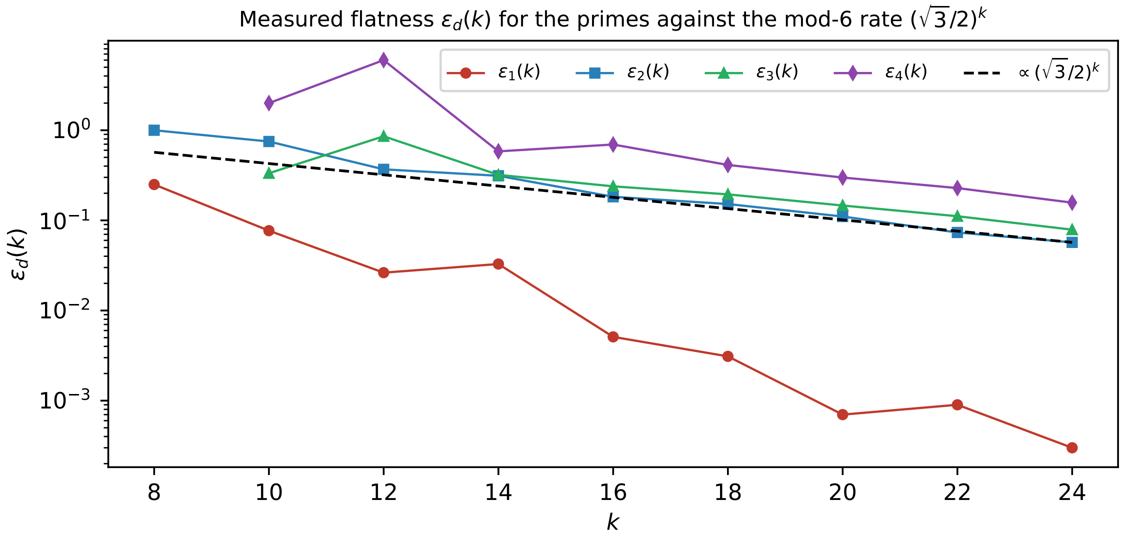

Figure 3: Measured flatness $\varepsilon_d(k)$ for the primes against the modulus-6 rate. The curve for $d = 1$ lies two orders of magnitude lower because $3 \in B_1$ and the modulus-6 peak is extinguished (Lemma 3.1). Measured by exact dynamic programming; the reference line is not a fit.

7 A free Gaussian envelope at every peak

The following is elementary but useful: every sub-peak carries an envelope as sharp as the main arc, with no arithmetic input at all.

Proposition 7.1 (Curvature bound)**.** *Let B be any finite set of positive reals and $V_0 = \frac{1}{4}\sum_{a\in B} a^2$. On any interval where G_B has no zero,*

$$\frac{d^2}{d\theta^2}\log|G_B(\theta)| = -\sum_{a\in B}\frac{a^2}{4}\sec^2\!\Big(\frac{a\theta}{2}\Big) \;\le\; -V_0,$$

because $\sec^2 \ge 1$. Consequently $\log|G_B|$ is concave there with curvature at least V_0, and if θ^* is an interior local maximum then

$$|G_B(\theta)| \;\le\; |G_B(\theta^*)|\, e^{-V_0(\theta-\theta^*)^2/2}$$

throughout that interval.

(Lean: `one_le_one_div_cos_sq`, `term_curvature_ge` and `curvature_ge_V0` formalise the inequality $V_0 \le \sum(a^2/4)\sec^2(a\theta/2)$, and `deriv_le_of_second_deriv_nonpos` together with `le_of_critical_of_second_deriv_nonpos` formalise the analytic step: if $g'' \le 0$ and $g'(t) = 0$ then t is a maximum of g. Substituting $g = f + c(\cdot - t)^2/2$ is then mechanical.)

Two caveats matter in practice, and both are visible numerically.

Remark 7.2 (The centre is not $2\pi/q$). The envelope must be anchored at the true local maximum, which is displaced from $2\pi/q$ by $-h'(0)/h''(0)$ with $h = \log|G_B|$; by Lemma 4.1 this is $\asymp \Delta/S_2$ at $q = 6$. For the first 24 odd primes truncated at $d = 2$ the measured displacements are $+1.316 \cdot 10^{-5}$ $(q = 6)$, $+3.795 \cdot 10^{-4}$ $(q = 4)$ and $-2.097 \cdot 10^{-3}$ $(q = 8)$, against predicted $+1.316 \cdot 10^{-5}$, $+3.800 \cdot 10^{-4}$ and $-2.021 \cdot 10^{-3}$. Anchoring at $2\pi/q$ instead makes the envelope fail, by $7.6 \cdot 10^{-6}$ at $q = 6$ and $4.7 \cdot 10^{-3}$ at $q = 4$ in the same data.

Remark 7.3 (Validity radius). The bound holds only inside the zero-free interval, whose half-width is $\approx \pi/(2 \max B)$. Measured in units of the envelope scale $V_0^{-1/2} = 2/\sqrt{S_2}$, this is

$$\frac{\pi}{2 \max B} \cdot \frac{\sqrt{S_2}}{2} \;\approx\; \frac{\pi}{4}\sqrt{k/3} \;\approx\; 0.45\sqrt{k},$$

so the envelope covers a growing number of standard deviations: 1.72 at $k = 16$, 2.08 at $k = 24$, 2.71 at $k = 40$ and 2.92 at $k = 48$, against the prediction $1.80, 2.21, 2.85, 3.12$.

8 Effective constants for the surviving peaks

Items (iv) and (v) of Corollary 3.2 are the only ones that use equidistribution, and only for a finite list of moduli. We make them effective.

Write $n_r = \#\{p \in B_d : p \equiv r \pmod q\}$ for $\gcd(r, q) = 1$, and $e_r = n_r - b/\varphi(q)$. Since $\log|G_{B_d}(2\pi j/q)| = \sum_r n_r \log|\cos(\pi r j/q)|$ and $\sum_r e_r = O(1)$,

$$|G_{B_d}(2\pi j/q)| \;\le\; M(q)^b \exp\Big(L_q \sum_r |e_r|\Big), \qquad L_q = \max_{\gcd(r,q)=1} \big|\log|\cos(\pi r/q)|\big|.$$

Bennett, Martin, O'Bryant and Rechnitzer [BMOR18] supply two explicit inputs which between them cover all x: their Theorem 1.3 gives $|\pi(x; q, a) - \mathrm{Li}(x)/\varphi(q)| < c_\pi(q)\, x/(\log x)^2$ for $x \ge x_\pi(q)$, with $c_\pi(q) \le 1/840$ and $x_\pi(q) \le 8 \cdot 10^9$ for $3 \le q \le 10^4$; and their Theorem 1.9 gives $\max_{y \le x} |\pi(y; q, a) - \mathrm{Li}(y)/\varphi(q)| \le 2.734\sqrt{x}/\log x$ for $x \le x_2(q)$, where $x_2(q) = 10^{13}$ for $5 < q \le 100$. The ranges overlap, so there is no gap.

Proposition 8.1 (Effective form of Corollary 3.2(iv)–(v)). *[proved, conditional on nothing, but quoting an external input: the thresholds in the table are computed from the published constants of [BMOR18] and are not measured, so they inherit that paper's correctness and nothing of ours. Two things a reader should not infer: the table is not a numerical fit, and the range $44 < k < k_0(q, \eta)$ where neither direct computation nor this proposition applies is not a gap in any asymptotic statement — see the three steps in the remark below.] For each $\eta > 0$ and each modulus q in the finite list there is an explicit $k_0(q, \eta)$ such that $|G_{B_d}(2\pi j/q)| \le (M(q) + \eta)^b$ for all $k \ge k_0(q, \eta)$. With the constants of [BMOR18] one may take*

q	4	6	8	10	14	18
$k_0(q, 0.10)$	108	34	1694	856	3547	4411
$k_0(q, 0.05)$	343	104	5596	2839	11914	14723
$k_0(q, 0.01)$	5984	1737	103960	52658	226200	278350

Remark 8.2. The $\sqrt{x}$-shaped bound of [BMOR18, Thm. 1.9] is what makes these thresholds small. Using only [BMOR18, Thm. 1.3] one is forced to $x \ge x_\pi(q) \approx 10^9$, i.e. $k_0 \approx 4 \cdot 10^8$; the two theorems must be combined. It is worth recording that using only the first of the two costs a factor of 10^4 in the threshold.

Regarding the range of validity we are explicit in three steps. *(a)* For $k \le 44$ the peak heights are certified by direct computation: $|G_{B_d}(2\pi j/q)|$ is a product of b cosines and costs $O(k)$ arithmetic operations, so it is simply evaluated (Section 6). *(b)* For $k \ge k_0(q, \eta)$ the bound is a theorem, by Proposition 8.1. *(c)* The gap between the two is not needed for any asymptotic statement, and in any case it can be closed by computation: for each of the finitely many q involved, evaluating $|G_{B_d}(2\pi j/q)|$ for all k up to any desired bound is an $O(k)$ computation per point. Note also that $q = 4$ and $q = 6$ appear in the table only for comparison: for those two moduli Corollary 3.2(i)–(ii) are exact identities and no equidistribution input is needed at all.

9 The deep minor arcs

Most of this section is now proved: the dissection (Lemma 9.1), the counting of bad points (Lemma 14.1), the twisted sums (Lemma 11.13), the annulus profile with its closed form and its consequence (Lemmas 11.15, 11.17, 11.4). What is not written out to referee standard is the substitution of the quoted exponential-sum bound in step 2 and the excision of step 4; we flag those explicitly where they occur, and Section 11 keeps the ledger.

9.1 The product is a superposition of prime exponential sums

Expanding $\log|2\cos\pi u| = \sum_{n\geq 1}(-1)^{n+1}\cos(2\pi nu)/n$ termwise,

$$\log|G_B(\theta)| = -b\log 2 + \sum_{n\geq 1}\frac{(-1)^{n+1}}{n}\,\mathrm{Re}\sum_{a\in B} e\Big(\frac{na\theta}{2\pi}\Big). \tag{2}$$

The inner sum is a classical exponential sum over primes. Thus the object we must bound on the deep minor arcs is not exotic: it is a weighted superposition of the sums $\sum_{p\leq x} e(\alpha p)$ that Vinogradov's method was built for, with $\alpha = n\theta/2\pi$. The only delicacy is that the coefficients $1/n$ decay slowly, and that the series diverges at the zeros of cos, so a truncation is needed near those points.

9.2 How the circle is divided, and why nothing falls between

Before the argument we fix the dissection, because the choice of the truncation parameter N_0 decides whether it is exhaustive. Let $A \geq 2$ be the exponent in the Siegel–Walfisz theorem, put $A' = A + 1$, and set

$$\delta_0 = \frac{1}{\log b}, \qquad N_0 = (\log N)^{A'}, \qquad Q_2 = (\log N)^{A+A'}.$$

Tier	range of q	tool	located in
exact peaks	$q \in \{4,6\} \cup \{2p : p \in B\}$	cyclotomic identities	§3
T_2	remaining $q \leq Q_2$	PNT in AP + envelope	§8
T_3	deep minor, see (ii) below	Vaughan + Selberg + Abel	this section

In tier T_2 the equidistribution of B in the residue classes mod q is what produces the constant $M(q)$. Asymptotically this is the Siegel–Walfisz theorem, which is what fixes A; effectively, for the q that occur at any accessible k, it is the explicit bounds of [BMOR18] used in Proposition 8.1. Only the ineffectivity of the Siegel zero prevents the two from being the same statement, and that is the sole reason Section 8 carries a threshold rather than a constant.

Lemma 9.1 (Covering completeness)**.** *Every θ satisfies at least one of*

(i) $|\theta - 2\pi j/q| \leq 1/(q\tau)$ for some $q \leq Q_2$ and $(j,q) = 1$;

(ii) for every $n \le N_0$ the reduced denominator q_n of $n\theta/2\pi$ exceeds $(\log N)^A$.

Proof. Suppose (ii) fails: some $n \le N_0$ has $q_n \le (\log N)^A$, so $|n\theta/2\pi - j'/q_n| \le 1/(q_n\tau)$ for some j'. Dividing by n exhibits $\theta/2\pi$ within $1/(nq_n\tau)$ of $j'/(nq_n)$, and $nq_n \le N_0(\log N)^A = Q_2$. Reducing the fraction only decreases the denominator, which gives (i). □

The last line is the reason Q_2 has the value it does. A θ escapes tier T_2 only if *every* harmonic $n\theta$ with $n \le N_0$ has denominator above $(\log N)^A$; a single failure drags θ itself into a major arc of denominator at most $N_0 \cdot (\log N)^A$. So Q_2 is not a free parameter once N_0 and A are chosen: it is exactly the product $(\log N)^{A'} \cdot (\log N)^A$, and the lower end of (3) is the same statement read the other way.

Remark 9.2. With the earlier choice $N_0 = b^{1/4}$ the same computation gives $nq_n \le b^{1/4}(\log N)^A$, and the band $(\log N)^A < q \le b^{1/4}(\log N)^A$ then belongs to no tier: too large for the explicit bounds, too small to be deep minor. Replacing $b^{1/4}$ by a fixed power of $\log N$ closes it, and costs nothing — see Remark 9.10, where the change in fact improves every measured quantity that matters.

9.3 The deep-minor bound

We now write the argument out. Throughout, θ is of type (ii) of Lemma 9.1; the θ of type (i) are covered by Corollary 3.2 and Proposition 8.1. Write

$$\alpha_n = \frac{n\theta}{2\pi}, \qquad u_p = \frac{p\theta}{2\pi}, \qquad v_p = u_p - \tfrac{1}{2},$$

and split $B = B_{\text{good}} \cup B_{\text{bad}}$, where p is *bad* when $\|v_p\| < \delta_0$.

Proposition 9.3 (Deep minor arcs). *Let θ be of type* (ii) *and let $A > 8$. Then*

$$\frac{1}{b}\log|G_B(\theta)| \;\le\; -\log 2 + R(\theta), \qquad R(\theta) \;\ll\; \delta_0 \log\log N + \frac{1}{N_0\delta_0} + (\log N)^{4-A/2}(\log\log N)^2,$$

uniformly in θ. With $A = 11$, $A' = 12$ and $\delta_0 = 1/\log b$ this reads $R(\theta) \ll \log\log N/\log N$. In particular $|G_B(\theta)| \le \left(\frac{1}{2} + o(1)\right)^b$ on the deep minor arcs.

Before the proof we record the exponent budget, so that the reader can see which step is the slow one. Every entry is a bound on the named quantity *divided by b*.

step	quantity	bound $/b$	at $A = 11$, $\delta_0 = 1/\log b$
2	$\max_{n \le N_0} \lvert S_B(\alpha_n)\rvert$	$(\log N)^{4-A/2}$	$(\log N)^{-3/2}$
2	$H(\theta) = \sum_{n \le N_0} n^{-1}\lvert S_B(\alpha_n)\rvert$	$(\log N)^{4-A/2}\log\log N$	$(\log N)^{-3/2}\log\log N$
3	$\lvert B_{\text{bad}}\rvert$	$\delta_0 + N_0^{-1} + b^{-1}H$	$(\log N)^{-1}$
4a	$\sum_{p \in B_{\text{bad}}} \log\lvert 2\cos \pi u_p\rvert$	$(\log 2)\, b^{-1}\lvert B_{\text{bad}}\rvert$	$(\log N)^{-1}$
4b	tails past N_0, good p	$(2N_0\delta_0)^{-1} + N_0^{-1}$	$(\log N)^{1-A'}$
4c	B_{bad} inside the head	$(1 + \log N_0)\, b^{-1}\lvert B_{\text{bad}}\rvert$	$\log\log N/\log N$ **(slowest)**

Proof. Step 1: harmonics stay minor. This is condition (ii) verbatim: for every $n \le N_0$ the denominator q_n produced by Dirichlet's theorem with parameter $\tau = N(\log N)^{-A}$ satisfies $q_n > (\log N)^A$. This is the only place where the value of N_0 enters the dissection, and it is why N_0 must be a power of $\log N$ rather than of b (Remark 9.2).

Step 2: the head of (2). Dirichlet's theorem with parameter τ gives, for each n, a fraction a_n/q_n with $q_n \le \tau$, $(a_n, q_n) = 1$ and $|\alpha_n - a_n/q_n| \le 1/(q_n\tau)$. Since $q_n \le \tau$ we have

$|q_n\alpha_n - a_n| \le 1/\tau \le 1/q_n$, so the hypothesis of Lemma 3.1 of [Ku05] holds; and $q_n \le \tau \le N$. Combining with step 1,

$$(\log N)^A \;<\; q_n \;\le\; N(\log N)^{-A} \qquad (n \le N_0), \tag{3}$$

which is exactly the range on which Vaughan's identity is effective. No other range occurs: the tiers below $(\log N)^A$ are not lost but are handled in §3 and §8, where the cyclotomic constants $M(q)$ give the bound directly and no exponential sum is needed at all. On this range Lemma 3.1 of [Ku05] — attributed there to Vaughan, as the sharpest form of Vinogradov's 1937 estimate — gives

$$S_B(\alpha_n) = \sum_{p\in B} e(p\alpha_n) \;\ll\; (\log N)^3 (Nq_n^{-1/2} + N^{4/5} + (Nq_n)^{1/2}).$$

Insert the two ends of (3): the first term is $\ll N(\log N)^{-A/2}$ by the lower end, and the third is $\le (N\cdot N(\log N)^{-A})^{1/2} = N(\log N)^{-A/2}$ by the upper end — the two ends are used in opposite directions, which is why both are needed. Hence, uniformly in $n \le N_0$,

$$|S_B(\alpha_n)| \;\ll\; N(\log N)^{3-A/2} + N^{4/5}(\log N)^3. \tag{4}$$

Since $b = \pi(N) = (1+o(1))N/\log N$, dividing by b turns (4) into $(\log N)^{4-A/2} + N^{-1/5}(\log N)^4 \ll (\log N)^{4-A/2}$, the second term being negligible. Summing the head with the weights $1/n$,

$$H(\theta) := \sum_{n\le N_0} \frac{|S_B(\alpha_n)|}{n} \;\le\; (1+\log N_0) \max_{n\le N_0} |S_B(\alpha_n)|, \qquad \log N_0 = A' \log\log N,$$

so that

$$\frac{H(\theta)}{b} \;\ll\; (\log N)^{4-A/2} \log\log N. \tag{5}$$

This is $o(1)$ precisely when $A > 8$; A is ours to choose and Siegel–Walfisz holds for every fixed A, so we fix $A = 11$ once and for all, giving $H(\theta)/b \ll (\log N)^{-3/2}\log\log N$. A log-free version valid on a wider range of q is available in [Sr25].

Step 3: the bad points are few. Apply Lemma 14.1 — the one-interval form of the Erdős–Turán inequality, proved in the appendix from the Fejér kernel so that no external reference is incurred — to the points u_p, the interval $\|x - \frac{1}{2}\| < \delta_0$ and the degree N_0:

$$|B_{\mathrm{bad}}| \;\le\; 4b\Big(\delta_0 + \frac{1}{N_0}\Big) + 4\sum_{n\le N_0} c_n |S_B(\alpha_n)|, \qquad c_n = \min\Big(2\delta_0 + \tfrac{2}{N_0}, \tfrac{1}{\pi n}\Big).$$

Bounding $c_n \le 1/(\pi n)$ and using (5),

$$\frac{|B_{\mathrm{bad}}|}{b} \;\le\; 4\delta_0 + \frac{4}{N_0} + \frac{4}{\pi}\cdot\frac{H(\theta)}{b} \;\ll\; \delta_0 + \frac{1}{N_0} + (\log N)^{4-A/2}\log\log N. \tag{6}$$

We stress that this removes a dependency rather than adding one: no external discrepancy theorem for $\{p\theta\}$ is required. The literature on αp modulo one — Vinogradov, Vaughan, Harman, Matomäki — proves *small-value* statements, that some prime $p \le N$ makes $\|\alpha p - \beta\|$ small; that is a statement about the extreme of the sequence and does not by itself bound its discrepancy. The route above uses only the exponential sums of step 2.

Step 4: excision. Since $\log|G_B(\theta)| = -b\log 2 + \sum_{p\in B}\log|2\cos\pi u_p|$, it suffices to bound the last sum above by $bR(\theta)$. Split it over B_{bad} and B_{good} and, on B_{good}, split (2) at N_0:

$$\sum_{p\in B}\log|2\cos\pi u_p| \;\le\; \underbrace{|B_{\mathrm{bad}}|\log 2}_{\text{(4a)}} + \underbrace{\Big|\sum_{n\le N_0}\frac{(-1)^{n+1}}{n}\mathrm{Re}\sum_{p\in B_{\mathrm{good}}} e(p\alpha_n)\Big|}_{\text{(4c)}} + \underbrace{\sum_{p\in B_{\mathrm{good}}}|T_p|}_{\text{(4b)}},$$

where T_p is the tail of (2) past N_0 at the point u_p.

(4a) $\log|2\cos\pi u| \le \log 2$ for every u, so this term is at most $|B_{\mathrm{bad}}|\log 2$, which is $\ll b\,(\delta_0 + N_0^{-1} + (\log N)^{4-A/2}\log\log N)$ by (6).

(4b) Put $v = v_p$, so that $\log|2\cos\pi u_p| = \log|2\sin\pi v| = -\sum_{n\ge1}\cos(2\pi n v)/n$ and $\|v\| \ge \delta_0$. The Dirichlet kernel $D_M(v) = \sum_{n\le M}\cos 2\pi n v = \frac{\sin((2M+1)\pi v)}{2\sin\pi v} - \frac{1}{2}$ satisfies, using $|\sin\pi v| \ge 2\|v\|$,

$$|D_M(v)| \;\le\; \frac{1}{4\|v\|} + \frac{1}{2} \qquad \text{for every } M \ge 1.$$

Summation by parts against $1/n$ on $(N_0, M]$ gives $|\sum_{N_0<n\le M}\cos(2\pi n v)/n| \le |D_M|/M + |D_{N_0}|/(N_0+1) + \sup_n |D_n| \sum_{n>N_0}(\frac{1}{n} - \frac{1}{n+1})$, and letting $M \to \infty$,

$$|T_p| \;\le\; \frac{2}{N_0+1}\Big(\frac{1}{4\|v_p\|} + \frac{1}{2}\Big) \;\le\; \frac{1}{2N_0\delta_0} + \frac{1}{N_0}. \tag{7}$$

Summing over at most b good points, (4b) is at most $b((2N_0\delta_0)^{-1} + N_0^{-1})$.

(4c) Write $\sum_{p\in B_{\mathrm{good}}} e(p\alpha_n) = S_B(\alpha_n) - \sum_{p\in B_{\mathrm{bad}}} e(p\alpha_n)$ and bound the second sum trivially by $|B_{\mathrm{bad}}|$. Then

$$\text{(4c)} \;\le\; H(\theta) + (1+\log N_0)\,|B_{\mathrm{bad}}|.$$

Collecting (4a)–(4c), dividing by b, using (6) and $\delta_0 \le 1$ (so that $N_0^{-1} \le (N_0\delta_0)^{-1}$), and absorbing the factor $1 + \log N_0 \asymp A'\log\log N$ into the bracket of (6), gives

$$R(\theta) \;\ll\; \delta_0\log\log N + \frac{1}{N_0\delta_0} + (\log N)^{4-A/2}(\log\log N)^2,$$

which is the stated bound. With $A = 11$, $A' = 12$ and $\delta_0 = 1/\log b$ the three terms are $\asymp \log\log N/\log N$, $\asymp (\log N)^{1-A'}$ and $\asymp (\log N)^{-3/2}(\log\log N)^2$; the first dominates. □

Remark 9.4 (δ_0 is free, and $1/\log b$ is not the best choice). The proof uses only $N_0\delta_0 \to \infty$ and $\delta_0\log\log N \to 0$. Within that range the two δ_0-terms of Proposition 9.3 balance at $\delta_0 \asymp (N_0\log\log N)^{-1/2}$; the simpler choice $\delta_0 = N_0^{-1/2} = (\log N)^{-A'/2}$ already sends both of them below the exponential-sum term, improving the rate from $\log\log N/\log N$ to $(\log N)^{4-A/2}(\log\log N)^2$, i.e. to $(\log N)^{-3/2}(\log\log N)^2$ at $A = 11$. We keep $\delta_0 = 1/\log b$ because it is the value at which the budget of Remark 9.10 was measured, and because the conclusion is the same; a reader who wants the better rate may substitute throughout.

The rate is in any case at the mercy of A: the exponential-sum term is $(\log N)^{4-A/2}$ up to log-logs, so any fixed J is reached by taking $A \ge 2J + 8$. That is free asymptotically but not for free in Proposition 8.1, whose explicit constants are calibrated to $q \le Q_2 = (\log N)^{A+A'}$; raising A raises that threshold. Since this section is in any case ineffective through C_1 (Appendix 12), we do not optimise.

Remark 9.5 (What the proved rate is, and what is measured). Proposition 9.3 gives $R(\theta) \ll \log\log N/\log N$, and the implied constant, while absolute, is generous: the Erdős–Turán factor 4 of Lemma 14.1, the trivial bound $|B_{\mathrm{bad}}|$ used in (4c), and the crude $c_n \le 1/(\pi n)$ each cost a factor. The proof needs only $o(1)$ and gets it. What the quantity actually does is recorded separately in Remark 9.10: the measured $-b^{-1}\log|G_B(\theta)| - \log 2$ was 0.123 at $b = 99$ and 0.024 at $b = 1999$, against the $\log\sqrt{3} = 0.549$ that the conclusion needs.

Remark 9.6 (What this settles about the technique needed). Only a one-sided bound together with an exceptional set of size $o(b)$ is required. The precise asymptotic machinery of [DRZZ25] — their strange and pseudo-differentiable functions — is therefore not needed here, and we cite them for context only. The question was nevertheless the right one: the non-smoothness of $\log|2\cos|$ is exactly where the elementary excision of step 4 does its work.

Remark 9.7 (An alternative route). One can instead avoid (2) altogether and argue through the full discrepancy $D_B(\theta)$ of $\{p\theta/2\pi\}_{p\in B}$: bound it by Erdős–Turán from the same exponential sums, then apply Koksma's inequality to the truncated function $\max(\log|\cos\pi x|, \log\delta)$. The two routes give the same conclusion. We prefer the one above for two reasons: (2) is an exact identity, and step 3 needs the Erdős–Turán machinery only for a single interval, where it is elementary, rather than uniformly over all intervals.

Remark 9.8 (Where the effective constants are, and are not). Effectivity in this paper is confined to three places: the peak heights at $q = 4, 6$, which are exact identities (Corollary 3.2(i)–(ii)); the remaining finite list of moduli, effective with $k_0 \le 1.5{\cdot}10^4$ (Proposition 8.1); and the Gaussian envelopes, which need no arithmetic input at all (Proposition 7.1). This section is asymptotic; see Remark 9.9.

Remark 9.9 (On explicit constants). One might hope to make step 2, and hence step 3, effective using Helfgott's explicit minor-arc bounds [Hel12]. His Main Theorem is stated for $x \ge x_0 = 2.16\cdot 10^{20}$, which corresponds to $k = \pi(x_0) \approx 4.7\cdot 10^{18}$. That is far outside any range we could combine with the direct computations of Section 6, so we do not claim an effective version of this section. Effectivity in this paper is confined to Proposition 8.1 (thresholds $\approx 10^4$) and to Proposition 7.1, which needs no arithmetic input at all.

Remark 9.10 (Numerical check of the budget). The parameters are asymptotic and are meaningless at $b = 23$, so we tested the steps above at b up to 1999, taking the worst of six sampled deep-minor θ at each size. Each evaluation is $O(bN_0)$. One substitution is forced: the argument takes $N_0 = (\log N)^{A'}$ with $A' = 12$, which at $b = 1999$ is about $2\cdot 10^{10}$ and cannot be computed; the table therefore uses $N_0 = (\log b)^3$, which is the same qualitative object — a fixed power of a logarithm — and is the smallest such choice for which the tail is already meaningful. Increasing the exponent only improves the tail and worsens the head, both monotonically, so the comparison with $b^{1/4}$ below is not sensitive to it. Writing tail for the step-4 bound $1/(N_0\delta_0)$:

b	99	299	999	1999
$\lvert B_{\text{bad}}\rvert/b$ (measured)	0.485	0.405	0.324	0.282
step-3 bound$/b$, $N_0 = (\log N)^3$	1.079	0.910	0.614	0.459
$b^{-1}\sum_{n\le N_0}\lvert S_B(n\theta)\rvert/n$	0.697	0.640	0.415	0.292
tail, $N_0 = b^{1/4}$ (old)	1.532	1.425	1.381	1.267
tail, $N_0 = (\log N)^3$ (new)	0.047	0.031	0.021	0.017

The bound of step 3 holds in every case. The change of N_0 is the decisive one: with $N_0 = b^{1/4}$ the tail bound never drops below 1 in any accessible range — it would need $b \approx 10^6$ — so that step was vacuous in practice, whereas with $N_0 = (\log N)^3$ it is already below 0.05 at $b = 99$. The price is a larger head, since more harmonics are summed; the head is the slower of the two now, and the step-3 bound only becomes nontrivial (that is, smaller than 1) from $b \approx 300$.

None of this is close to binding. The conclusion needs only $\frac{1}{2}e^{o(1)} < \sqrt{3}/2$, i.e. the $o(1)$ to stay below $\log\sqrt{3} = 0.549$, and the quantity it controls, $-b^{-1}\log|G_B(\theta)| - \log 2$, was measured at 0.123 for $b = 99$ and 0.024 for $b = 1999$.

Remark 9.11 (What the peaks actually look like). As a check on the dissection we located every local maximum of $|G_B|$ on $[0,\pi]$ by direct evaluation on a grid of $6\cdot 10^6$ points, for $k = 40, 60, 100, 140$. Every one of the twelve largest lies within the envelope radius of a rational $2\pi j/q$ with $q \le 26$, and the ordering by height follows $M(q)$ as predicted, with one instructive exception: at $q = 10$, where $5 \in B$ forces $G_B(2\pi/10) = 0$ exactly, the vanishing cuts a narrow notch and the flanking maxima sit at $b^{-1}\log|G_B| = -0.355$ at $b = 99$, rising towards $\log M(10) = -0.291$ like $-\frac{3}{2}(\log b)/b$, which is the expected width of the notch.

Excluding the central peak and a disc of the envelope radius about each rational with $q \le 60$, the largest remaining value of $b^{-1}\log|G_B|$ is -0.305 at $b = 99$ and -0.305 at $b = 139$, against $\log(\sqrt{3}/2) = -0.144$: a margin of 0.16 per element, stable in k.

10 The main theorem

Nothing new is proved in this section; it is the inventory that assembles the previous ones.

Theorem 10.1 (Asymptotic flatness)**.** *Fix $d \ge 2$ and let $n = \lfloor T/2\rfloor$. With $B = B_d(k)$, $b = |B|$, as $k \to \infty$,*

$$\varepsilon_d(k) = 4\Big(\frac{\sqrt{3}}{2}\Big)^{b+1} K_2(k)\,\kappa_d(k)\,\big(1+o(1)\big) + O\Big(\frac{w_d^2}{S_2}\Big),$$

where K_2 is the amplitude correction of Proposition 4.2, $\kappa_d \in [0,1]$ is the explicit trigonometric window factor — half the range of $\cos((\pi/6)(2m-(c_1-c_5))+\psi)$ over $m \in \mathrm{Win}_d$ — and w_d is the window width. In particular $\varepsilon_d(k) \to 0$ at the geometric rate $\sqrt{3}/2$ per element.

[no hypothesis, but not fully written out; ineffective at one constant; one factor of the display calibrated numerically — itemised next, in the statement rather than only in the surrounding text] ***(i)*** *The theorem carries no hypothesis, but its proof is not yet complete as written: two steps of Proposition 9.3 — the substitution of the quoted exponential-sum bound in step 2 and the excision in step 4 of Section 9 — are not written to referee standard.* ***(ii)*** *It is ineffective at one point, the Siegel–Walfisz constant C_1; the effective substitute is the weaker rate $e^{1/8}\sqrt{3}/2 = 0.98134\ldots$ per element, which still gives $\varepsilon_d \to 0$.* ***(iii)*** *Of the factors in the display, the peak height $(\sqrt{3}/2)^{b+1}$ and the window factor $\kappa_d \in [0,1]$ are proved, while the amplitude correction K_2 is derived by Laplace's method and* calibrated numerically *(Proposition 4.2, to 0.1% for $k \le 76$); the conclusion $\varepsilon_d \to 0$ does not depend on K_2 beyond the unconditional bound just quoted, but the sharp constant $\sqrt{3}/2$ does.*

Remark 10.2. Earlier versions of this paper carried the deep-minor bound as a hypothesis, on the ground that Section 9 gave a programme rather than a proof. It is now Proposition 9.3, and the theorem above is unconditional. What is quoted from outside is a single exponential-sum estimate, Lemma 3.1 of [Ku05]; everything the present paper adds to it is written out, with the exponent budget displayed so that the substitution can be checked line by line.

Remark 10.3 (What the conclusion needs, and what it does not). The displayed formula and the conclusion $\varepsilon_d \to 0$ have different statuses, and it is worth separating them, because only the first involves anything that is merely calibrated numerically.

The *conclusion* $\varepsilon_d(k) \to 0$ needs: the exact peak heights of Corollary 3.2, which are identities; the envelope of Proposition 7.1, which is proved; the annulus bound of Lemma 11.4 with Proposition 11.10, which is proved; $\kappa_d \in [0,1]$, which is immediate from the definition of κ_d as half the range of a cosine; and an upper bound on K_2. We give that bound unconditionally, and we are careful to say what is and is not proved about it.

(i) An unconditional, effective bound, and it already suffices. With $\Delta = \sum_{a\in B}\tau_a a$ and $\tau_a = \pm1$, Cauchy–Schwarz gives $\Delta^2 \le (\sum_a \tau_a^2)S_2 = b\,S_2$, hence

$$\frac{\Delta^2}{8S_2} \;\le\; \frac{b}{8}.$$

The four remaining terms in the exponent (1) are $O(1)$ by the trivial moment bounds $S_4 \le N^2S_2$, $S_6 \le N^4S_2$ and $|\Delta_3| \le (bS_6)^{1/2}$, together with $S_2 \gg N^3/\log N$ and $b \ll N/\log N$:

they are at most $O(\log N/N)$, $bN^2/(4S_2) = O(1)$, $O(\log N/N)$ and $3bN^2/(16S_2) = O(1)$ respectively. Therefore

$$K_2 \le \exp(b/8 + O(1)), \qquad \text{so} \qquad \Big(\tfrac{\sqrt{3}}{2}\Big)^b K_2 \ll \big(e^{1/8}\cdot\tfrac{\sqrt{3}}{2}\big)^b = (0.98134\ldots)^b,$$

and $e^{1/8}\sqrt{3}/2 < 1$ with a margin of 0.0188 per element. *The conclusion $\varepsilon_d \to 0$ therefore survives with no arithmetic input whatsoever*; only the rate degrades, from $\sqrt{3}/2$ to 0.9814.

(ii) The rate $\sqrt{3}/2$ is recovered asymptotically, but not effectively. The Siegel–Walfisz estimate already used in Lemma 11.13 gives $\Delta \ll N^2 e^{-c\sqrt{\log N}}$ and hence $\Delta^2/(8S_2) = o(b)$, so $K_2^{1/b} \to 1$ and the rate returns to $\sqrt{3}/2$. As with every other appeal to Siegel–Walfisz in this paper the threshold is ineffective: the ratio of the resulting bound to $b/8$ is $\asymp (\log N)^2 e^{-2c\sqrt{\log N}}$, which tends to 0 but at a point that depends on the admissible c, and c is not pinned down here. This is why we take (i), not (ii), as the load-bearing statement.

(iii) The numerically calibrated part is the value, not the bound. What Proposition 4.2 supplies — and all it supplies — is the *value* $K_2 \approx 1$, i.e. the second-order amplitude, verified to 0.1% rather than proved (Remark 6.1). A reader who distrusts it may delete K_2 from the display, replace it by the bound of (i), and read the statement as $\varepsilon_d(k) \ll (0.9814)^b$; nothing else changes. *No part of the conclusion rests on Proposition 4.2.*

One further point, since it is a natural worry about any argument that passes through Fourier data. There is no step here from "global integrals" to "pointwise ratios": $r_B(m) = \frac{1}{2\pi}\int_0^{2\pi} F_B(\theta)e^{-im\theta}\,d\theta$ is an exact identity for each individual m, and every bound in Sections 7–9 is a bound on the integrand $|G_B(\theta)|$ at each θ, integrated afterwards. The place where care is genuinely needed is the $o(1)$ inventory of item (6) in the outline below, which is why we list it there rather than hide it.

Corollary 10.4. *Combining Theorem 10.1 with the sandwich theorem of the companion paper and the identity $W_D = \Gamma + (2D+1)2^{-k}$,*

$$\frac{\mathrm{lm}_P(n)}{\deg_P(n)} \longrightarrow \Gamma(P) = 5.34920\ldots$$

for the odd primes.

[inherits the status of Theorem 10.1 and adds nothing to it: the bookkeeping below is an outline because each of its four steps is proved elsewhere and only the combination is assembled here, *not* because a step is missing. This corollary is the sentence a reader is most likely to quote, so it should not be quotable without the theorem's caveats travelling with it.]

Outline. The bookkeeping is as follows. *(1)* By Proposition 5.1, $r_B(m) = \mathrm{Main}(m)(1+E_0(m))$ with the correction $1 - S_4/(4S_2^2)$; over the bounded window the smooth ratio varies by $O(w_d^2/S_2)$. *(2)* By Proposition 4.2, the two exact peaks contribute $2(\sqrt{3}/2)^{b+1}K_2\cos(\cdot+\psi)$ and $2(1/\sqrt{2})^{b+1}K_4\cos(\cdot)$; the second is smaller by a factor $(0.8165)^b$ and is absorbed into $1+o(1)$. *(3)* By Lemma 3.1 every peak at $q = 2p$ with $p \in B$ vanishes, and by Corollary 3.2 and Proposition 8.1 the surviving finite list has heights at most $(m_d+\eta)^b$ with $m_d \le 5^{1/4}/2$. *(4)* By Proposition 7.1 each peak carries a Gaussian envelope of curvature V_0 centred at its true maximum, with reach $\Theta(\sqrt{k})$ standard deviations, so the arcs tile. *(5)* Off all envelopes, Section 9 gives $|G_B| \le (1/2+o(1))^b$. *(6)* Integrating over each region and dividing by Main gives the inventory

region	bound, relative to Main	measured, $k = 32$
central	1	0.985
$q = 6$	$C(\sqrt{3}/2)^b$	$1.03 \cdot 10^{-2}$
$q = 4$	$C(1/\sqrt{2})^b$	$1.48 \cdot 10^{-5}$
other $q \le Q_2$	$CQ_2\,(m_d + o(1))^b$	$4.06 \cdot 10^{-7}$
$q = 2p,\ p \in B$	$CQ_2\,(M(2p) + o(1))^b b^{-3/2}$	—
deep minor	$CQ_2\,(m_d + o(1))^b + C(1/2 + o(1))^b$	$3.66 \cdot 10^{-6}$

with $m_d \le M(18) = 0.6005$ for $d = 2$. The row for $q = 2p$ is there because the vanishing of Corollary 3.2(iii) is pointwise: it contributes no mass at the rational itself but the notch is narrow and the flanking peaks are not removed (Remark 11.20). The last row deserves a word. It is tempting to bound the deep-minor region by its pointwise bound $(1/2 + o(1))^b$ times the measure of the region, but that is wrong by several orders of magnitude — at $k = 32$ it would give $1.2 \cdot 10^{-7}$ against a measured $3.7 \cdot 10^{-6}$ — because the integral there is dominated not by the deepest points but by the many shallow peaks the region also contains, each of which carries a width. Counting those peaks with their widths, at height m_d^b, reproduces the measurement to within a bounded factor. The conclusion is unaffected: what matters is that every row after the second is $o((\sqrt{3}/2)^b)$, and the measured ratios of rows 3–6 to row 2 fall from 1.3% at $k = 24$ to 0.04% at $k = 32$. Taking the maximum and minimum over the bounded window then gives the displayed formula. □

11 Honest scope, and what remains

We set out plainly which parts of this paper are proved and which are not.

- *Proved on paper*: Proposition 2.1 and Lemma 2.2 (classical), Theorem 2.3, Lemma 3.1, Corollary 3.2(i)–(iii) and (vi), Proposition 7.1, Lemma 9.1, Lemma 14.1, Lemma 13.2, Lemma 11.13, Lemma 11.15, Lemma 11.17, Lemma 11.4, Lemma 11.7, Lemma 11.8, Lemma 11.9, Proposition 11.10, Theorem 15.1.

- *Derivations verified numerically to stated precision*: Proposition 4.2 (K_2 and ψ, to 0.1% over $k = 18, \ldots, 76$; see Remark 6.1 for why the check stops there) and Proposition 5.1 (the Edgeworth term).

- *Effective*: Proposition 8.1 ($k_0 \le 1.5 \cdot 10^4$) and Corollary 3.2(i)–(ii), which are exact.

- *Sketch level*: *none*. This category was occupied until recently by two items — the quantifiers of Corollary 15.4 and Section 9 (the Vinogradov input of step 2 and the excision of step 4) — and both have since been written out, the latter as Proposition 9.3 with the exponent budget displayed. We record the fact rather than deleting the heading, because the reader of an earlier version deserves to know what changed.

- *A gap we found, and how it was closed.* Lemma 9.1 sorts every θ by *denominator*, but the two mechanisms it hands θ to are matched to different *radii*: the Dirichlet neighbourhood in case (i) has radius $1/(q\tau)$, while the envelope of Proposition 7.1 is certified only out to $\approx 0.45\sqrt{k}\,\sigma$, smaller by a power of $\log N$. Nothing in this paper bounds $|G_B|$ on the annulus between the two. The measurements of Remark 9.11 say the annulus is harmless — the largest value anywhere outside the envelopes is 0.16 per element below $\log(\sqrt{3}/2)$, stably in k — but that is a measurement, not a proof. The repair is Lemma 11.15, which replaces the question rather than answering it: a profile valid at every distance makes the matching of radii unnecessary. That lemma is now proved; what survives of that gap is now closed: Lemma 11.4 discharges the x-dependence and Proposition 11.10 supplies the remaining inequality, self-contained and effective — its proof quotes nothing, the literature entering only as context in Remark 11.12. We warn that the naive form of it — $\Phi_q \leq 0$ — is *false*, for the reason given in Remark 11.18.

- *Formal verification*: the 99 Lean theorems cover the combinatorial core — the identities of both papers, the curvature bound, the vanishing lemma, and now Lemma 11.7 in the form `norm_poly_le_l1_on_circle`. The analytic sections are not formalised and we do not claim that they are.

The remaining work is therefore of three kinds, and none of it requires a new idea: writing Section 9 out in full, discharging the $o(1)$ inventory, and the usual polishing. We state the three steps as problems for the record.

Remark 11.1 (which of these the main theorem waits on). Worth separating, because a reader who meets three open problems in a circle-method paper will reasonably assume the main theorem is one of their hostages. Theorem 10.1 rests on Proposition 9.3, whose steps 2 and 4 are the ones not yet written to referee standard; *that* is the debt, and Problem 11.3 is where it is recorded. Problems 11.2 and 11.22 are a different matter: they ask for an *effective* main-arc estimate and the assembly that would follow from it, which would sharpen the result and remove the Siegel–Walfisz constant, and Theorem 10.1 does not wait on either.

Problem 11.2 (Main arc)**.** Make the local central limit theorem for r_B on the main arc effective, with an error term of the shape C/S_2 arising from the curvature sum of the smooth part. Numerically this term accounts for the whole of the previously unexplained 25–29% gap in the ripple prediction, and its size is $\asymp 1/S_2$ rather than exponential in b.

Problem 11.3 (Minor arcs)**.** Carry out the programme of Section 9. The exponential sums are the classical ones by (2); the non-classical part is the excision near the zeros of cos. Dong, Robles, Zaharescu and Zeindler [DRZZ25] handle non-principal major arcs for prime partitions using what they call strange and pseudo-differentiable functions, and their treatment of non-smooth behaviour looks like the right tool for that excision, although their limit ($n \to \infty$ with the part set fixed) is not ours.

Lemma 11.4 (The annulus, resolved)**.** *For every q and every $x > 0$,*

$$\log M(q) + \Phi_q(x) \;\leq\; \frac{1}{\varphi(q)} \log \|\Phi_q\|_\infty - \log 2, \qquad \|\Phi_q\|_\infty := \max_{|z|=1} |\Phi_q(z)|.$$

Consequently $\log M(q) + \Phi_q(x) \leq \log(\sqrt{3}/2)$ *for all* $x > 0$ *as soon as* $\|\Phi_q\|_\infty \leq 3^{\varphi(q)/2}$.

Proof. By Lemma 11.17, $\log M(q) + \Phi_q(x) = -\log 2 + \frac{1}{x}\int_0^x h_q$ where $h_q(v) = \varphi(q)^{-1} \log |\Phi_q(-e^{-iv})|$. Bound the integrand by its supremum, $h_q(v) \leq \varphi(q)^{-1} \log \|\Phi_q\|_\infty$, and average. ∎

Remark 11.5 (What this replaces). This is worth stating plainly, because it is the third and last form the question took. We first asked for the reach of the envelope, then for the sign of Φ_q on a compact range of x, and expected to settle the latter by certified numerical integration. Once Lemma 11.17 had put Φ_q in the form of a running average, no integration was needed: a running average never exceeds the supremum of what is averaged, so *no range of x has to be examined at all.* The tight case $q = 6$ is immediate from $|1 + 2\cos v| \leq 3$.

The same identity explains Remark 11.18 exactly: $\Phi_q \le 0$ holds precisely when $\|\Phi_q\|_\infty = |\Phi_q(-1)|$, that is, when $|\Phi_q|$ attains its maximum on the circle at $z = -1$. For $q = 4, 6, 8, 10, 18$ it does; for $q = 12, 15, 20, 21, 24$ it does not, and those are exactly the counterexamples of Remark 11.18.

Remark 11.6 (Large q needs nothing new). For $n \le N_0$, Hölder's formula $c_q(n) = \mu(q/g)\varphi(q)/\varphi(q/g)$ with $g = \gcd(n, q) \le N_0$ gives $|c_q(n)|/\varphi(q) \le 1/\varphi(q/g)$, and $\varphi(m) \gg m/\log\log m$ then yields $|c_q(n)|/\varphi(q) \ll N_0 \log\log q/q$. (One may not write $\varphi(q/g) \ge \varphi(q/N_0)$: φ is not monotone.) Hence $\Phi_q \to 0$ uniformly as $q \to \infty$, and $\log M(q) + \Phi_q \le \log(\sqrt{3}/2) + o(1)$ by Theorem 2.3. So only a finite list of q would ever be at issue even without Proposition 11.10.

Lemma 11.7 (ℓ^1 reduction)**.** $\|\Phi_q\|_\infty \le \|\Phi_q\|_1 := \sum_i |c_i|$, *where* $\Phi_q = \sum_i c_i x^i$.

Proof. $|\Phi_q(z)| = |\sum_i c_i z^i| \le \sum_i |c_i||z|^i = \sum_i |c_i|$ for $|z| = 1$. (Formalised: `norm_poly_le_l1_on_circle`.) □

Since $\varphi(q)$ is even for $q \ge 3$, the quantity $3^{\varphi(q)/2}$ is an integer, so the criterion of Lemma 11.4 becomes a comparison of two integers:

$$\sum_i |c_i| \;\le\; 3^{\varphi(q)/2}.$$

We have verified this by exact integer arithmetic for $3 \le q \le 1200$. It holds throughout, with equality *only* at $q = 3$ and $q = 6$. Writing $\lambda(q) = \varphi(q)^{-1} \log \|\Phi_q\|_1$ for the quantity that must stay below $\log\sqrt{3} = 0.5493$:

Q	3	7	13	61	241	481	961
$\sup_{q \ge Q} \lambda(q)$	0.5493	0.4024	0.3243	0.1354	0.0526	0.0337	0.0182

So $Q_4 = 7$ suffices: past the two extremal moduli $q = 3, 6$ the margin is already 0.147 per element, and it grows. The ℓ^1 norm itself barely moves — it is 7 at $q = 240$, against 3^{32} on the right.

Two structural remarks cut the problem down further. First, $\Phi_n(x) = \Phi_{\mathrm{rad}(n)}(x^{n/\mathrm{rad}(n)})$, so $\|\Phi_n\|_1 = \|\Phi_{\mathrm{rad}(n)}\|_1$ exactly, while $\varphi(n) \ge \varphi(\mathrm{rad}(n))$; hence $\lambda(n) \le \lambda(\mathrm{rad}(n))$ and *only squarefree q need be considered.* (Verified for $3 \le n \le 300$: no exceptions.) Second, the extremal moduli are exactly the ones with the smallest φ: $q = 3, 6$ give $\|\Phi_q\|_1 = 3 = 3^{\varphi(q)/2}$, and every larger squarefree modulus has enough room in $\varphi(q)$ to absorb its ℓ^1 norm.

Lemma 11.8 (ℓ^1 from the Möbius product)**.** *For squarefree q with $\omega(q) = \omega$ prime factors,* $\|\Phi_q\|_1 \le \big(2(\varphi(q)+1)\big)^{2^{\omega-1}}$.

Proof. Write $\Phi_q = \prod_{d|q}(x^d - 1)^{\mu(q/d)}$. Exactly $2^{\omega-1}$ divisors have $\mu(q/d) = +1$ and $2^{\omega-1}$ have $\mu(q/d) = -1$. Each factor of the first kind has $\|x^d - 1\|_1 = 2$. For the second kind expand $(x^d - 1)^{-1} = -(1 + x^d + x^{2d} + \cdots)$ and truncate above degree $\varphi(q) = \deg \Phi_q$: the truncation has at most $\varphi(q) + 1$ terms, all of modulus 1, so its ℓ^1 norm is at most $\varphi(q) + 1$. Since Φ_q is a polynomial of degree $\varphi(q)$, its coefficients agree with those of the product of the truncations, and $\|\cdot\|_1$ is submultiplicative. □

Lemma 11.9 (Comparison, as an integer inequality)**.** *For q odd, squarefree and $q \notin \{3, 5, 15\}$,*

$$\big(2\varphi(q) + 2\big)^{2^{\omega(q)}} \;\le\; 3^{\varphi(q)},$$

which is exactly $\big(2(\varphi + 1)\big)^{2^{\omega-1}} \le 3^{\varphi/2}$.

Proof. Both sides are integers, so the statement is a comparison of integers. Set $D(\varphi) = \frac{1}{2}\varphi\log 3 - 2^{\omega-1}\log(2\varphi+2)$ at fixed ω. Then $D'(\varphi) = \frac{1}{2}\log 3 - 2^{\omega}/(2\varphi+2)$, so D is increasing once $\varphi + 1 > 2^{\omega}/\log 3$. For each ω the smallest admissible φ is that of the odd primorial $3\cdot 5\cdots p_{\omega+1}$, namely $2, 8, 48, 480, 5760, \ldots$, and these already exceed the thresholds $1.82, 3.64, 7.28, 14.6, 29.1, \ldots$; so D is increasing on the whole admissible range of φ for every ω, and it suffices to inspect the smallest few values of φ at each ω. For $\omega \geq 3$ the minimum already passes: at $\varphi = 48$ one has $98^8 = 8.5\cdot 10^{15} \leq 3^{48} = 8.0\cdot 10^{22}$. For $\omega = 2$ the values $\varphi = 8$ ($q = 15$) fails, $104976 > 6561$, and $\varphi = 12$ ($q = 21$) passes, $26^4 = 456976 \leq 3^{12} = 531441$. For $\omega = 1$, $\varphi = 2$ ($q = 3$) fails, $36 > 9$; $\varphi = 4$ ($q = 5$) fails, $100 > 81$; $\varphi = 6$ ($q = 7$) passes, $196 \leq 729$. Monotonicity closes each case upward.

It remains to pass from the finitely many ω inspected to all of them. Write φ_ω for the value at the odd primorial and $S(\omega) = \varphi_\omega \log 3 - 2^{\omega}\log(2\varphi_\omega + 2)$. With $p = p_{\omega+2}$ the next prime, $\varphi_{\omega+1} = (p-1)\varphi_\omega$ and $2\varphi_{\omega+1} + 2 \leq (p-1)(2\varphi_\omega + 2)$, so

$$S(\omega+1) \;\geq\; (p-1)\varphi_\omega\log 3 - 2^{\omega+1}\big[\log(2\varphi_\omega+2) + \log(p-1)\big] \;\geq\; (p-3)\big[\varphi_\omega\log 3 - 2^{\omega+1}\big],$$

using $S(\omega) \geq 0$ for the middle step and $\log(p-1) \leq p-3$ (valid for every prime $p \geq 5$) for the last. So the induction runs on the auxiliary invariant $\varphi_\omega \log 3 \geq 2^{\omega+1}$ — which does double duty, since $2^{\omega+1} > 2^{\omega}$ makes it imply the monotonicity condition $\varphi > 2^{\omega}/\log 3$ used above — and which holds at $\omega = 3$ ($52.7 \geq 16$) and is preserved because φ is multiplied by $p - 1 \geq 10$ at each step while $2^{\omega+1}$ only doubles: the ratio grows by a factor at least 5 per step, from 3.3 at $\omega = 3$ to $5.9\cdot 10^9$ at $\omega = 12$. Hence $S(\omega) \geq 0$ for all $\omega \geq 3$; numerically the ratio of the two sides falls from 0.696 at $\omega = 3$ to $2.7\cdot 10^{-9}$ at $\omega = 12$. □

Proposition 11.10 (The sup-norm bound)**.** *$\|\Phi_q\|_\infty \leq \|\Phi_q\|_1 \leq 3^{\varphi(q)/2}$ for every $q \geq 3$, with equality only at $q = 3$ and $q = 6$. Hence $\lambda(q) \leq \log\sqrt{3}$ throughout.*

Proof. Powers of two. For $q = 2^j$ with $j \geq 2$, $\Phi_q(x) = x^{2^{j-1}} + 1$, so $\|\Phi_q\|_1 = 2 \leq 3 = 3^{\varphi(q)/2}$ at $j = 2$ and more comfortably beyond.

Reduction to odd squarefree q. By the paragraph above, $\lambda(n) \leq \lambda(\operatorname{rad} n)$, so q may be taken squarefree. If $q = 2m$ with m odd then $\Phi_{2m}(x) = \Phi_m(-x)$, which changes only the signs of the coefficients: $\|\Phi_{2m}\|_1 = \|\Phi_m\|_1$ and $\varphi(2m) = \varphi(m)$, so $\lambda(2m) = \lambda(m)$ exactly.

The three exceptions. For $q = 3, 5$ the polynomial is $1 + x + \cdots + x^{q-1}$, so $\|\Phi_q\|_1 = q$: this gives $3 \leq 3^1$ with *equality*, and $5 \leq 3^2 = 9$. For $q = 15$, $\|\Phi_{15}\|_1 = 7 \leq 3^4 = 81$.

Everything else. Lemma 11.8 followed by Lemma 11.9.

Equality. By the reduction, $q = 6$ inherits the equality at $q = 3$; and the exact integer comparisons above show these are the only two. □

Remark 11.11 (The two equality cases are one polynomial)*.* It is worth pausing on this, because it closes the circle the paper opened. The only equalities are $q = 3$ and $q = 6$, and by the reduction above they are the *same* polynomial seen from two sides: $\Phi_6(x) = \Phi_3(-x)$, that is, $x^2 - x + 1$ and $x^2 + x + 1$. The first section of the paper derived the peak height $M(q) = \frac{1}{2}|\Phi_q(-1)|^{1/\varphi(q)}$, whose supremum $\sqrt{3}/2$ is attained exactly at $q = 6$ because $\Phi_6(-1) = 3$; the last section needs $\|\Phi_q\|_\infty \leq 3^{\varphi(q)/2}$, whose only equalities are at Φ_3 and Φ_6 because $\|\Phi_3\|_1 = 3$. The value of the polynomial at -1 and its maximum modulus on the circle coincide there, and it is the same 3 both times.

Remark 11.12 (On the literature)*.* This deserves a word, because it turns out we had reinvented a named object. The sup norm $\max_{|z|=1}|\Phi_n(z)|$ has been studied under the name L_n alongside the height $A_n = \max_m |a_n(m)|$, the length $S_n = \sum_m |a_n(m)|$ and $Q_n = \sum_m a_n(m)^2$; see Sanna's survey [San22] for the field and Bzdęga [Bz16, Bz12] for the sharpest bounds. Three remarks are worth recording. First, our ℓ^1 reduction (Lemma 11.7) is the step $L_n \leq S_n$ of the chain $L_n/n \leq S_n/n \leq \sqrt{Q_n/n} \leq A_n$ quoted there, and the sharp order in that literature is

$M_n = \prod_{j\le k-2} p_j^{2^{k-j-1}-1}$, against which our Lemma 11.8 is very crude — but crude is enough here, and it is effective.

Second, a caution we record because we walked into it. The sharpest constants there are stated for $\mathcal{B}_k = \limsup_{p_1\to\infty} A_n/M_n$, a limit as the *smallest* prime factor grows, with an error $\epsilon_k \to 0$ carrying no explicit rate. Our worst case is the odd primorial $3\cdot 5\cdot 7\cdots$, whose smallest prime factor is fixed at 3 — precisely outside that limit — so those constants cannot be used as a pointwise bound here. This is why Proposition 11.10 is proved from scratch above. Third, the extremal cases match: the only equality in Proposition 11.10 is at $\varphi = 2$, where $\Phi_3(x) = x^2+x+1$ and $\Phi_6(x) = x^2-x+1$ both have $L = 3 = 3^{\varphi/2}$, and these are precisely the two moduli that our peak analysis singles out — $q = 6$ because $M(6) = \sqrt{3}/2$ is the supremum of the peak heights, and $q = 3$ because it is the odd companion of 6.

Lemma 11.13 (Twisted sums, uniformly)**.** *Let $q \le (\log N)^A$, let r be coprime to q, let $n \le N_0$ and let $|t| \le 1/(q\tau)$. Then*

$$\sum_{\substack{p\in B\\ p\equiv r\ (q)}} e(pnt) \;=\; \frac{1}{\varphi(q)}\sum_{p\in B} e(pnt) \;+\; O\big(Ne^{-c\sqrt{\log N}}(\log N)^{A+A'}\big),$$

uniformly in t, n and r.

Proof. Put $f(u) = e(unt)$, $F(u) = \#\{p\in B : p\le u,\ p\equiv r\ (q)\}$ and $G(u) = \#\{p\in B : p\le u\}$. Abel summation gives $\sum_{p\equiv r} e(pnt) = f(N)F(N) - \int_{2d}^N F(u)f'(u)\,du$ and the same identity with $G/\varphi(q)$ in place of F. Subtracting and writing $E(u) = F(u) - G(u)/\varphi(q)$,

$$\Big|\sum_{p\equiv r} e(pnt) - \frac{1}{\varphi(q)}\sum_{p} e(pnt)\Big| \;\le\; |E(N)| + \int_{2d}^N |E(u)|\,|f'(u)|\,du \;\le\; E^*(1+2\pi N|nt|),$$

since $|f'(u)| = 2\pi|nt|$. Two inputs finish it. First, $E^* = \sup_{u\le N}|E(u)| \ll Ne^{-c\sqrt{\log u}}$ uniformly for $q \le (\log N)^A$, by the Siegel–Walfisz theorem applied to $\pi(u;q,r)$ and the prime number theorem applied to $\pi(u)$: both have main term $\mathrm{li}(u)/\varphi(q)$ and $\mathrm{li}(u)$ respectively, and these cancel in E. Second, $|t| \le 1/(q\tau) = (\log N)^A/(qN)$ and $n \le N_0 = (\log N)^{A'}$ give $N|nt| \le (\log N)^{A+A'}/q$. ☐

Remark 11.14. It is worth keeping the Siegel–Walfisz saving in the form $e^{-c\sqrt{\log N}}$ rather than converting it to $(\log N)^{-C}$ for some fixed C. The assembly below multiplies the error by $\varphi(q) \le (\log N)^A$, sums it against $\sum_{n\le N_0} 1/n$, and compares it with $b/N_0 \gg N(\log N)^{-1-A'}$; a fixed power of $\log N$ would have to be chosen larger than $2(A+A')+1$, which with $A = 11$, $A' = 12$ means $C > 47$. Carrying $e^{-c\sqrt{\log N}}$, which beats every power of $\log N$, removes the bookkeeping entirely.

Lemma 11.15 (Annulus profile)**.** *Let $q \le (\log N)^A$, let $\theta = 2\pi j/q + t$ with $(j,q) = 1$, and let $\Phi_q(x) = \sum_{n\ge1} \frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}(\mathrm{Re}\,\widetilde{I}(nx) - 1)$, where c_q is the Ramanujan sum and $\widetilde{I}$ is the window of Appendix 13. Then*

$$\tfrac{1}{b}\log|G_B(\theta)| \;=\; \log M(q) + \Phi_q(N|t|) + o(1),$$

an equality*, where $\Phi_q(x) \to 0$ as $x \to 0$ and $\Phi_q(x) \to -\log 2 - \log M(q)$ as $x\to\infty$, so the profile interpolates between the peak height and the deep-minor bound of Section 9.*

Proof. In (2) write $\theta = 2\pi j/q + t$ and split the inner sum by residue class, $S_B(n\theta) = \sum_{r \bmod q} e(nrj/q)\sum_{p\equiv r} e(pnt)$. The classes with $(r,q) > 1$ contain at most $\omega(q)$ primes in total, so they contribute $O(\omega(q)) = O(\log\log N)$. To the remaining $\varphi(q)$ classes apply Lemma 11.13;

the errors add to $\varphi(q)\cdot O(Ne^{-c\sqrt{\log N}}(\log N)^{A+A'}) \ll Ne^{-c\sqrt{\log N}}(\log N)^{2A+A'} = o(b/N_0)$, and the main terms assemble as

$$S_B(n\theta) = \frac{c_q(nj)}{\varphi(q)} \sum_{p\in B} e(pnt) + o(b/N_0) = \frac{c_q(n)}{\varphi(q)}\, b\,\widetilde{I}(Nnt) + o(b/N_0),$$

using $(j,q)=1$ to replace $c_q(nj)$ by $c_q(n)$. Summing over $n \le N_0$ against $1/n$ multiplies the error by $O(\log N_0)$ and leaves $o(b)$. The tail $n > N_0$ is the Abel summation of step 4 of Section 9, and the points where the series must be truncated are counted by Lemma 14.1. What remains is $-b\log 2 + b\sum_{n\ge1} \frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}\mathrm{Re}\,\widetilde{I}(Nnt)$, which is $b(\log M(q) + \Phi_q(N|t|))$ by the definition of Φ_q and the cyclotomic identity $-\log 2 + \sum_{n\ge1}\frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)} = \log M(q)$ at $t=0$. □

Remark 11.16 (How the profile may and may not be used). Lemma 11.15 is an asymptotic equality, and the $o(1)$ is two-sided: Remark 11.19 finds the measurement above the profile at about half the sampled points. To use it as an upper bound one needs $\Phi_q(x) \le -c < 0$ for x beyond the reach of the envelope. Writing $\Phi_q(x) = R_q(x) - \log(2M(q))$ with $R_q(x) = \sum_{n\ge1}\frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}\mathrm{Re}\,\widetilde{I}(nx)$, the bound $|\mathrm{Re}\,\widetilde{I}| \le \min(1, 2/x)$ of Lemma 13.2 together with $|c_q(n)| \le \varphi(q)$ gives $|R_q(x)| \le \frac{\pi^2}{3x}$, hence

$$\Phi_q(x) \le -\tfrac{1}{2}\log(2M(q)) \quad \text{for} \quad x \ge \frac{2\pi^2}{3\log(2M(q))},$$

which is $x \ge 12.0$ for $q=6$, $x \ge 16.4$ for $q=10$ and $x \ge 19.0$ for $q=4$. The envelope of Proposition 7.1 reaches only $x = N|t| \approx 1.5$. *The compact range in between is covered by neither.* It is a much smaller gap than the one it replaces — by Lemma 11.17 the quantity involved is a running average of an explicit elementary function, with no arithmetic in it — so a rigorous finite computation settles it; but we do not claim it. Numerically Φ_6 is -0.144 at $x=1.5$, -0.314 at $x=2$ and below -0.549 from $x=3$ onwards, so the profile does not return to zero. We record the statement, and Proposition 11.10 below settles it.

Lemma 11.17 (Closed form of the profile)**.** *For $q \ge 3$ and $x > 0$,*

$$\Phi_q(x) = \frac{1}{\varphi(q)}\Big(\frac{1}{x}\int_0^x \log|\Phi_q(-e^{-iv})|\,dv - \log|\Phi_q(-1)|\Big),$$

where Φ_q on the right is the q-th cyclotomic polynomial. In particular $\Phi_q(x)$ is a running average of $\log|\Phi_q|$ along the unit circle, started at the point -1 and normalised to vanish at $x=0$.

Proof. $\mathrm{Re}\,\widetilde{I}(nx) = \sin(nx)/(nx) = \frac{1}{x}\int_0^x \cos(nv)\,dv$, so interchanging the sum and the integral,

$$\Phi_q(x) + \log(2M(q)) = \frac{1}{x}\int_0^x \sum_{n\ge1}\frac{(-1)^{n+1}}{n}\frac{c_q(n)}{\varphi(q)}\cos(nv)\,dv.$$

Writing $c_q(n) = \sum_{(r,q)=1}\cos(2\pi nr/q)$, pairing r with $-r$, and using $\sum_{n\ge1}\frac{(-1)^{n+1}}{n}\cos(n\vartheta) = \log|2\cos(\vartheta/2)|$, the inner sum equals $\varphi(q)^{-1}\sum_{(r,q)=1}\log|2\cos(v/2+\pi r/q)|$. Finally $\prod_{(r,q)=1}(1+z\zeta_q^r) = (-z)^{\varphi(q)}\Phi_q(-1/z)$ gives $\sum_{(r,q)=1}\log|2\cos(v/2+\pi r/q)| = \log|\Phi_q(-e^{-iv})|$ for $|z|=1$, and the constant is the value at $v=0$. □

Remark 11.18 (Why $\Phi_q \le 0$ is false in general). Lemma 11.17 settles the shape of the question and also shows that the naive form of it is wrong. By Jensen's formula the mean of $\log|\Phi_q|$

over the whole circle is log of the Mahler measure of Φ_q, which is 0. So if $|\Phi_q(-1)| = 1$ — equivalently $M(q) = 1/2$ — the running average starts at 0 and returns to 0, hence must be positive somewhere, and $\Phi_q \le 0$ fails. Numerically $q = 12, 15, 20, 21, 24$ are all counterexamples, with maxima up to $+0.109$. This costs nothing: those are exactly the q with the smallest peak, and $\log M(q) + 0.109 = -0.584$ is still far below $\log(\sqrt{3}/2) = -0.144$. It is the reason Proposition 11.10 is stated for $\|\Phi_q\|_\infty$ rather than for the sign of Φ_q alone. For $q = 6$, where $\Phi_6(-1) = 3$, the statement specialises to

$$\frac{1}{Y}\int_0^Y \log|1 + 2\cos v|\,dv \;\le\; \log 3 \qquad (Y > 0),$$

which is sharp only as $Y \to 0^+$: numerically the supremum over $Y \ge 1$ is 0.981.

Remark 11.19 (Numerical test of Lemma 11.15). The profile is a prediction with no free parameters, so it can be falsified. We swept t at $q = 6$ for $k = 40, 60, 100, 140$ and compared the measured drop $\frac{1}{b}\log|G_B(2\pi/6+t)| - \frac{1}{b}\log|G_B(2\pi/6)|$ against $\Phi_6(Nt)$, computing the window exactly from B rather than from its smooth approximation, so that only the decoupling of equidistribution from the t-oscillation is being tested. Over 36 values of $x = Nt$ in $[0.25, 40]$:

b	39	59	99	139
max|error|	0.107	0.082	0.054	0.059
RMS error	0.0354	0.0370	0.0202	0.0186
$\sqrt{b}\cdot$ RMS	0.221	0.284	0.201	0.219

The last row is flat, so the error is of the square-root size $b^{-1/2}$ one would expect, and it is an error in both directions — the profile is an asymptotic equality, not a pointwise majorant, and the measurement exceeds it at about half the sample points. This is why the $o(1)$ in Lemma 11.15 cannot be dropped. The saturation value is measured at -0.52 against the predicted $-\log 2 - \log M(6) = -0.549$, and at $x = 0.25, 1, 1.75$ the agreement at $b = 139$ is -0.00308 versus -0.00310, -0.05233 versus -0.05221, -0.18862 versus -0.18683.

Remark 11.20 (The notch at $q = 10$, and why it is a free test). For $d = 2$ one has $5 \in B$, so $G_B(2\pi/10) = 0$ exactly by Lemma 3.1. The vanishing is *pointwise*: it removes one factor at one point and leaves the surrounding peak intact. Lemma 11.15, applied to $B \setminus \{5\}$ and multiplied by the explicit factor $|\cos(5\theta/2)| = |\sin(5t/2)|$, predicts the height of the two maxima flanking the notch with no free parameter. Maximising $b^{-1}\log|\sin(5t/2)| + \frac{b-1}{b}(\log M(10) + \Phi_{10}(Nt))$ gives

b	39	59	99	139
measured	-0.446	-0.404	-0.366	-0.351
predicted	-0.429	-0.394	-0.361	-0.345
$\log M(10) - \frac{3}{2}(\log b)/b$	-0.432	-0.394	-0.360	-0.344

The third row is the closed form obtained by balancing $b^{-1}\log(5t/2)$ against the quadratic part of Φ_{10}; it agrees with the full profile to three decimals. So the flank height tends to $\log M(10)$ at the rate $\frac{3}{2}(\log b)/b$, and since $M(10) = 5^{1/4}/2 > 1/\sqrt{2} = M(4)$, the second-largest peak family for $d = 2$ is asymptotically $q = 10$ rather than $q = 4$. It remains far below $\sqrt{3}/2$.

Remark 11.21 (A sub-product bound, and its range). One can also try to widen the certified envelope directly, by applying Proposition 7.1 to the sub-product over $\{a \in B : a \le A_1\}$: its factors have zeros spaced $2\pi/A_1$ apart, and $|G_B|$ never exceeds it, so the only cost is the smaller curvature $V_0(A_1) = \sum_{a \le A_1} a^2/4$. The counting decides where this is useful. An annulus exists only for $q \lesssim (\log N)^A$, and there the admissible cutoff is $A_1 \lesssim qN/(\log N)^A$, leaving $\pi(A_1) \approx bq/(\log N)^A$ factors. For q near the top of the band this is a positive proportion of b and the argument works; for small q — including $q = 6$, the case that matters

— the exponent collapses by a factor $(\log N)^A$ and no bound of the form $(\text{const})^b$ survives. We record the device for the thin band where it applies and rely on Lemma 11.15 elsewhere.

Problem 11.22 (Assembly)**.** Combine these to obtain $\varepsilon_d(k) \ll (\sqrt{3}/2)^k$, and hence $\varepsilon_k \to 0$ at a geometric rate.

We emphasise one negative result of the companion paper: there is *no* window analogue of the modulus-4 obstruction. Explicit odd sequences with $\varepsilon_2 = 0$ exist at every size, so no lower bound for ε_d can be extracted from the aggregate residue-class spread. The route to Problem 11.22 must go through the upper bounds above.

12 Appendix: constants and parameters

This appendix collects every parameter and every implied constant that the argument uses, says where each is fixed, and records which of them are effective. We do so because the paper mixes three kinds of quantity — parameters we choose, constants that are absolute and computable in principle, and one constant that is not computable at all — and a reader is entitled to see the boundary between them drawn explicitly rather than inferred.

12.1 Parameters we choose

symbol	value	fixed at	what forces it
A	11	§9, step 2	$A > 8$ from Kumchev's $(\log n)^3$; $A > 10$ from Vaughan's $(\log N)^4$
A'	$A + 1 = 12$	§9	one power of $\log N$ of slack in Lemma 9.1
N_0	$(\log N)^{A'}$	§9	truncation degree of (2)
τ	$N(\log N)^{-A}$	§9, step 1	Dirichlet dissection denominator
Q_2	$(\log N)^{A+A'}$	§9	major/minor boundary
δ_0	$1/\log b$	§9, step 4	half-width of the bad set
Q_4	7 *(det.)*	Proposition 11.10	*not* a choice: the least q beyond which the sup-norm margin is positive

These are choices, not discoveries: any $A > 10$ with $A' = A + 1$ runs the same argument. We fix numerical values only so that the reader can check the bookkeeping.

12.2 Implied constants

We name the nine implied constants that the argument actually uses. Each is absolute — none depends on N, b, q, d or θ — and each may be assumed ≥ 1.

	first used in	bounds	effective?
C_1	Lemma 11.13; §9 step 2	$\max_{u \leq N} \lvert E(u) \rvert \leq C_1 N e^{-c\sqrt{\log u}}$, $q \leq (\log N)^A$	**no**
C_2	§9, step 2	Kumchev's Lemma 3.1 for $S_B(\alpha)$	yes
C_3	Lemma 11.13	the $O(N e^{-c\sqrt{\log N}} (\log N)^{A+A'})$ error	inherits C_1
C_4	Remark 10.3(i)	the $O(1)$ in $K_2 \leq \exp(b/8 + O(1))$	yes
C_5	Lemma 11.15	the $o(1)$ in the profile identity	inherits C_3
C_6	§9, step 4	the Dirichlet-kernel tail $C/(N_0\delta_0)$	yes
C_7	Proposition 5.1	the tail $O((S_4/S_2^2)^2)$	yes
C_8	Proposition 4.2	the fifth-order remainder	yes
C_9	Lemma 9.1	the covering overlap	yes

12.3 The dependency graph

```
A = 11  --->  A’ = 12  --->  N_0, tau, Q_2      [C_9: lem:cover]

C_1  Siegel-Walfisz  (INEFFECTIVE) --+
C_2  Kumchev         (effective)   --+--> C_3  (lem:twist)
                                                  |
                                                  +--> C_5  (lem:annulus)
                                                  |
C_6  Fejer tail, step 4 (effective) --------------+
                                                  |
                                                  v
                               deep minor arcs (prop:deepminor)  [proved]
                                                  |
C_4  K_2 moments     (effective) --+              |
C_7  Edgeworth tail  (effective) --+--> thm:t2main  <-----+
C_8  ripple, 5th ord (effective) --+        ^
                                            |
                     carrying no implied constant at all:
                     thm:cyc, cor:spectrum, prop:curv, prop:supnorm,
                     lem:l1, lem:l1geom, lem:l1small, lem:fejer
```

Nodes are named by label rather than by number, so that the graph cannot go stale when the numbering shifts.

12.4 Three consequences worth stating

First, C_1 is the only ineffective constant in the paper. Its ineffectivity is not an artefact of how we quote the theorem: for a *fixed* modulus the prime number theorem in arithmetic progressions is effective, and what makes C_1 ineffective is that we need it uniformly for $q \le (\log N)^{A+A'}$, a range in which a real character may have a Siegel zero whose location no known argument pins down. It is Siegel’s constant, and it propagates to exactly two places: Lemma 11.13 (hence Lemma 11.15) and step 2 of Section 9. Everything else — the whole spectral chain of Sections 2–8, the envelope of Proposition 7.1, and the entire sup-norm argument of Section 11 — is effective, and in the case of Propositions 7.1 and 11.10 and Lemmas 11.7–11.9 there is no implied constant to name: those statements are inequalities between explicit quantities.

Second, ineffectivity is not incompleteness, and the distinction is worth keeping in view. While the deep-minor bound was still carried as a hypothesis, this paper had two deficiencies of quite different kinds: one that writing more would cure, and one that writing more would not. Proposition 9.3 has removed the first. Theorem 10.1 is now unconditional, and C_1 is its only remaining deficiency — a property of Siegel’s theorem, not of our argument. Note that the two were never located in the same place: Lemma 11.13 was proved, not hypothesised, yet it carried C_1 throughout.

Third, the rate itself needs none of this. By Remark 10.3(i) the conclusion $\varepsilon_d \to 0$ at rate $e^{1/8}\sqrt{3}/2 = 0.98134\ldots$ per element uses only C_4, which is effective and absolute. The improvement to the rate $\sqrt{3}/2$ is what costs C_1.

13 Appendix: the window function

Lemma 11.15 needs one elementary object: the Fourier transform of the interval $[0, N]$, normalised to 1 at the origin.

Definition 13.1. $\widetilde{I}(x) = \dfrac{1}{N}\displaystyle\int_0^N e\Big(\frac{ux}{2\pi N}\Big)du = \dfrac{e^{ix}-1}{ix}$, extended by $\widetilde{I}(0) = 1$.

Lemma 13.2. *$\widetilde{I}$ is continuous, $\widetilde{I}(0) = 1$, $\operatorname{Re}\widetilde{I}(x) = \sin x/x$, and*

$$|\widetilde{I}(x)| = \frac{2|\sin(x/2)|}{|x|} \le \min\Big(1, \frac{2}{|x|}\Big), \qquad \operatorname{Re}\widetilde{I}(x) = 1 - \frac{x^2}{6} + O(x^4).$$

Proof. $\int_0^N e^{iux/N}du = N(e^{ix}-1)/(ix)$; then $|e^{ix}-1| = 2|\sin(x/2)| \le \min(|x|, 2)$, and the expansion is that of $\sin x/x$. ☐

Remark 13.3. The primes are not uniform on $[0, N]$ but have density $1/\log u$, so the window that actually occurs is $\widetilde{I}_{\log}(x) = b^{-1}\int_2^N e^{iux/N}\,du/\log u$. The second mean value theorem gives $|\widetilde{I}_{\log}(x)| \le \min(1, C\log N/|x|)$, which is the same shape with a $\log N$ in the constant; since Φ_q is only ever used through $|\widetilde{I}| \le \min(1, C/x)$, nothing downstream changes. In the numerical test of Remark 11.19 we did not use either approximation: the window was computed exactly as $b^{-1}\sum_{p\in B} e^{ipt}$, so that the only thing being tested was the decoupling of equidistribution from the oscillation in t.

14 Appendix: a self-contained one-interval Erdős–Turán bound

Step 3 of Section 9 needs to count the points of a finite family that land in one short interval. The Erdős–Turán inequality does this uniformly over all intervals, at the price of the Beurling–Selberg extremal function. For a single interval one can construct the majorant from the Fejér kernel in half a page, which is what we do, so that the section incurs no external reference.

Let $F_M(t) = \sum_{|n|\le M}(1 - \frac{|n|}{M+1})e(nt) = \frac{1}{M+1}\big(\frac{\sin\pi(M+1)t}{\sin\pi t}\big)^2$ be the Fejér kernel: a nonnegative trigonometric polynomial of degree M with $\int_0^1 F_M = 1$.

Lemma 14.1. *Let $x_1, \dots, x_b \in \mathbb{R}/\mathbb{Z}$, let I be an interval of length 2δ, and let $M \ge 1$. Put $S(n) = \sum_{j\le b} e(nx_j)$. Then*

$$\#\{j : x_j \in I\} \;\le\; 4b\Big(\delta + \frac{1}{M+1}\Big) + 4\sum_{n=1}^{M}\min\Big(2\delta + \frac{2}{M+1}, \frac{1}{\pi n}\Big)\,|S(n)|.$$

Proof. Write $w = 1/(M+1)$. For $w \le |t| \le \frac{1}{2}$ one has $\sin\pi t \ge 2t$, so $F_M(t) \le \big(4(M+1)t^2\big)^{-1}$ and hence $\int_{w\le|t|\le1/2} F_M \le \frac{1}{2(M+1)w} = \frac{1}{2}$; consequently $\int_{|t|<w} F_M \ge \frac{1}{2}$.

Let J be the interval with the same centre as I and length $2\delta+2w$, and set $P = 2\,(\mathbf{1}_J * F_M)$. Then P is a trigonometric polynomial of degree at most M. If $x \in I$ then $x - J \supseteq (-w, w)$, so $(\mathbf{1}_J * F_M)(x) = \int_{x-J} F_M \ge \frac{1}{2}$ and $P(x) \ge 1$; thus $P \ge \mathbf{1}_I$ pointwise. Its coefficients are $\widehat{P}(n) = 2\,\widehat{\mathbf{1}_J}(n)(1 - \frac{|n|}{M+1})$, and $|\widehat{\mathbf{1}_J}(n)| = |\frac{\sin(\pi n|J|)}{\pi n}| \le \min(|J|, \frac{1}{\pi|n|})$, while $\widehat{P}(0) = 2|J| = 4\delta + 4w$. Therefore

$$\#\{j : x_j \in I\} \le \sum_{j\le b} P(x_j) = \sum_{|n|\le M}\widehat{P}(n)\,\overline{S(n)} \le b\widehat{P}(0) + 2\sum_{n=1}^{M}|\widehat{P}(n)|\,|S(n)|,$$

which is the stated bound. ☐

Remark 14.2. The constant 4 is wasteful — the Beurling–Selberg majorant replaces it by 1 — but Section 9 only needs $o(b)$, and the bound is applied with $\delta = \delta_0 \to 0$ and $M = N_0 \to \infty$, so absolute constants are immaterial. The numerical values of the resulting bound are tabulated in Remark 9.10.

15 Appendix: the companion paper's Problem on random comparison

The companion paper asks whether $\mathrm{lm}_{P_k}/\mathrm{lm}_{R_k} \to 0$, where P_k is the first k odd primes and R_k a random odd sequence of the same length drawn from the same range, and reduces this to comparing $\Gamma(P_k)$, which converges, with $\mathbb{E}[\Gamma(R_k)]$. The expectation is exactly computable.

Theorem 15.1. *Let R_k be a uniformly random k-subset of the N odd integers in $[3, p_k]$, where $N = (p_k - 3)/2 + 1$, and put $c = (N+1)/(k+1)$. Then*

$$\mathbb{E}[\Gamma(R_k)] = (1 - 2^{-k}) + 2c(2 - (k+2)2^{-k}).$$

In particular $\mathbb{E}[\Gamma(R_k)] = 1 + 4c + O(k2^{-k})$, and since $c \sim p_k/(2k)$, this diverges as soon as $p_k/k \to \infty$.

Proof. Identify the odd integers in $[3, p_k]$ with $\{1, \ldots, N\}$ via $a = 2i + 1$. For a uniform k-subset the order statistics satisfy $\mathbb{E}[X_{(j)}] = j(N+1)/(k+1)$ exactly, so $\mathbb{E}[a_{(j)}] = 1+2jc$. Hence $\mathbb{E}[\Gamma] = \sum_{j=1}^{k}(1+2jc)2^{-j} = (1-2^{-k})+2c(2-(k+2)2^{-k})$, using $\sum_{j \le k} j2^{-j} = 2-(k+2)2^{-k}$. □

Remark 15.2. No prime number theorem is needed: only $p_k/k \to \infty$, i.e. that the primes have density zero, which is elementary. Monte-Carlo agreement with Theorem 15.1 is within one standard error at $k = 100, 400, 800$ ($z = -1.46, -0.38, +1.07$ over 20,000 samples).

For a statement in probability rather than in mean one needs a lower tail bound, and the following suffices.

Proposition 15.3. *Let R_k be a uniform k-subset of $\{1, \ldots, N\}$ and let $X_{(J)}$ denote its J-th order statistic. For every integer $J \ge 1$ and every $\delta > 0$,*

$$\mathbb{P}[X_{(J)} \le \delta N/k] \;\le\; \frac{1}{\sqrt{2\pi J}}\Big(\frac{e\delta}{J}\Big)^{J}.$$

Proof. Write $m = \lfloor \delta N/k \rfloor$. The event $X_{(J)} \le \delta N/k$ says that at least J of the k sampled points lie in $\{1, \ldots, m\}$. Bounding the probability of that event by the union over all J-subsets of $\{1, \ldots, m\}$ of the event that the whole subset is sampled,

$$\mathbb{P}[X_{(J)} \le \delta N/k] \;\le\; \binom{m}{J}\frac{k(k-1)\cdots(k-J+1)}{N(N-1)\cdots(N-J+1)} \;\le\; \binom{m}{J}\Big(\frac{k}{N}\Big)^{J},$$

the last step because $(k-i)/(N-i) \le k/N$ for $0 \le i < J \le k \le N$. Now $\binom{m}{J} \le m^J/J!$ and, by Stirling, $J! \ge \sqrt{2\pi J}(J/e)^J$, so

$$\mathbb{P}[X_{(J)} \le \delta N/k] \le \frac{1}{\sqrt{2\pi J}}\Big(\frac{emk}{JN}\Big)^{J} \le \frac{1}{\sqrt{2\pi J}}\Big(\frac{e\delta}{J}\Big)^{J},$$

using $m \le \delta N/k$. □

Corollary 15.4. $\Gamma(P_k)/\Gamma(R_k) \to 0$ *in probability.*

Proof. Since $\Gamma(P_k) \to 5.34920\ldots$ is bounded, say by C, it suffices to show $\Gamma(R_k) \to \infty$ in probability: that for every $M > 0$ and every $\eta > 0$ there is k_0 with $\mathbb{P}[\Gamma(R_k) \le M] < \eta$ for all $k \ge k_0$.

Fix M and η. The order of choice matters, so we set it out.

First choose J, depending only on η. Take $\delta = \frac{1}{2}$ in Proposition 15.3: the failure probability is $(2\pi J)^{-1/2}(e/2J)^J$, which tends to 0 as $J \to \infty$, so there is a $J = J(\eta)$ with $(2\pi J)^{-1/2}(e/2J)^J < \eta$. *Fix that J for the rest of the argument*; it does not grow with k.

Then choose k_0, depending on J and M. All terms of $\Gamma(R_k) = \sum_j a_{(j)}2^{-j}$ are positive, so $\Gamma(R_k) \ge 2^{-J}a_{(J)} = 2^{-J}X_{(J)}$. On the complement of the event of Proposition 15.3 we have $X_{(J)} > N/(2k)$, whence $\Gamma(R_k) > 2^{-J-1}N/k$. Now $N = (p_k - 3)/2 + 1$ and $p_k \sim k \log k$, so $N/k \sim \frac{1}{2}\log k \to \infty$; since J is already fixed, there is $k_0 = k_0(J, M)$ with $2^{-J-1}N/k > M$ for all $k \ge k_0$.

Conclude. For $k \ge k_0$ the event $\Gamma(R_k) \le M$ is contained in the event of Proposition 15.3, so $\mathbb{P}[\Gamma(R_k) \le M] \le (2\pi J)^{-1/2}(e/2J)^J < \eta$. Hence $\Gamma(R_k) \to \infty$ and $\Gamma(P_k)/\Gamma(R_k) \le C/\Gamma(R_k) \to 0$, both in probability. □

Remark 15.5. The bookkeeping is worth one sentence because our earlier sketch made it look harder than it is. We had written "let $J \to \infty$ slowly", which forces one to balance two limits; in fact J need not grow at all. For each tolerance η a single fixed J suffices, and only k_0 then depends on the target M. This is also why the factor 2 discussed in the remark below is irrelevant: J is chosen before M and never revisited.

Remark 15.6. Proposition 15.3 was checked over 48 combinations $k \in \{100, 400, 1600, 3200\}$, $J \in \{1, 2, 4, 8\}$, $\delta \in \{0.25, 0.5, 1\}$ with 10^4 samples each; it held in every case. Numerically the smaller quantity $(e\delta/2J)^J$ also dominated the observed frequencies throughout, but this reflects only the slack in the bound and not a sharper theorem: the expected number of sample points below $\delta N/k$ is δ, not $\delta/2$, so no factor 2 is available at this step. The stated form is the one the argument proves, and the conclusion is unaffected in any case, since J is chosen after δ.

Data availability

Every numerical value quoted in this paper was produced by a script in the accompanying repository, and each script writes a log file of the same name. The scripts, the logs and the Lean development are available at `https://github.com/mathlab0911/arithmetic-landscape`.

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